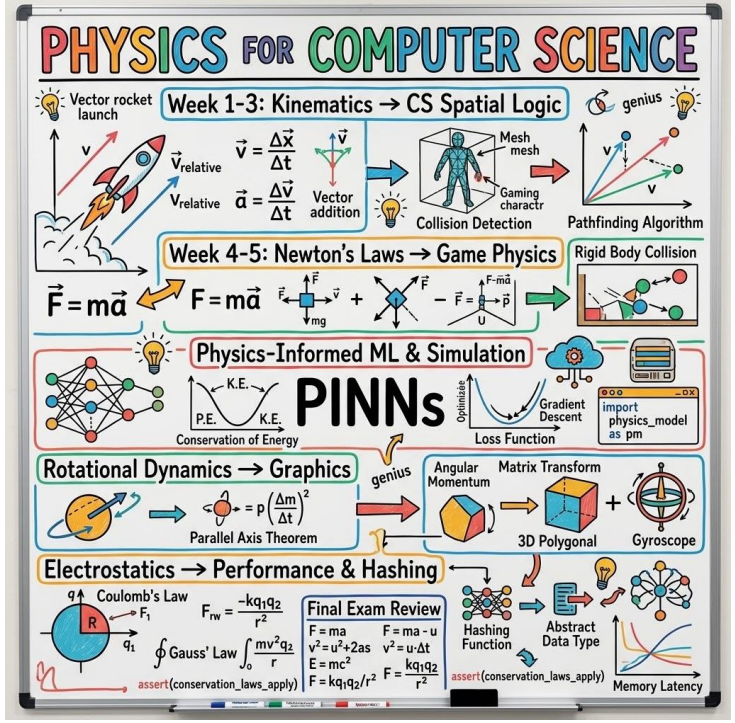
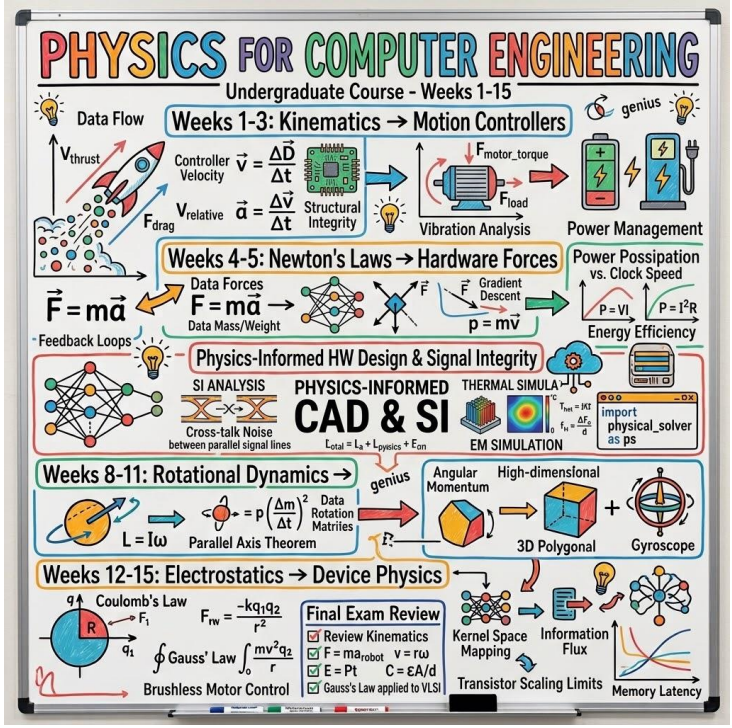
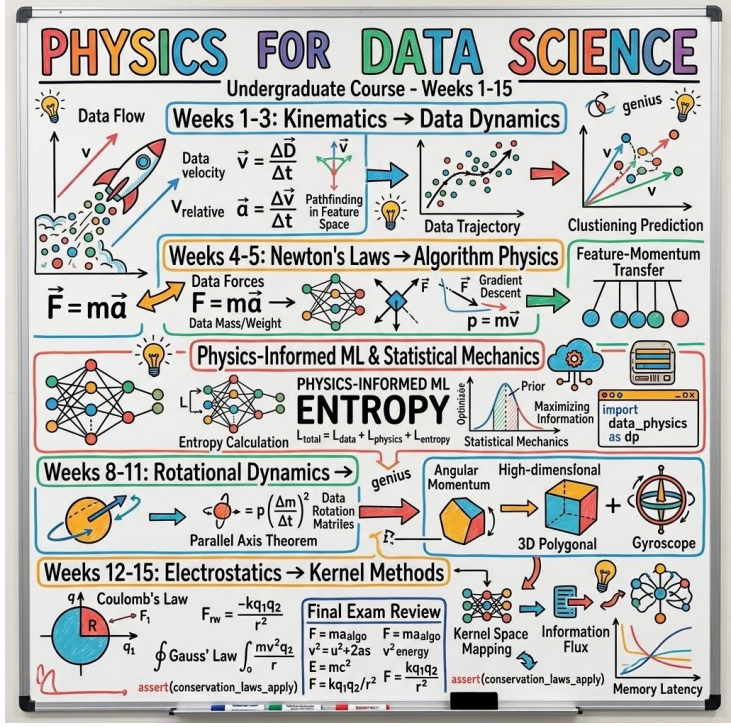
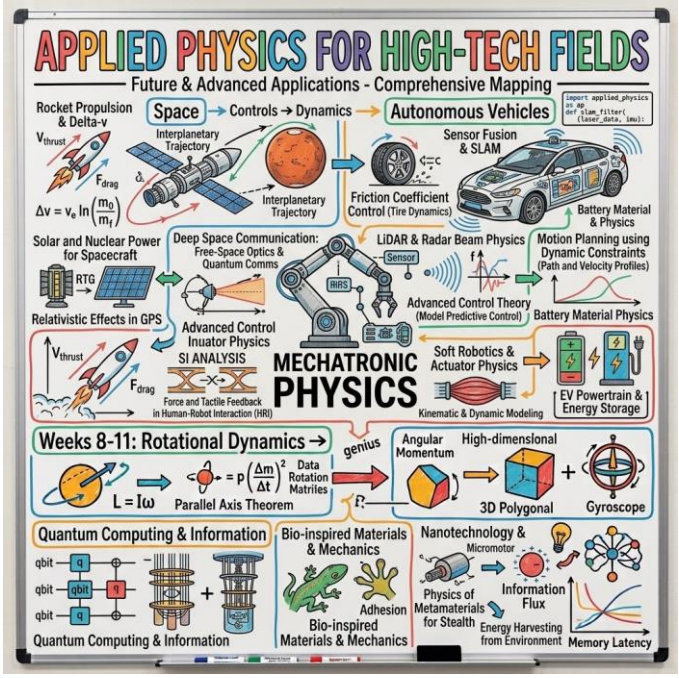








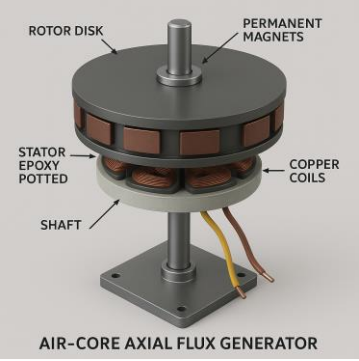
Applied Physics
PH101

Faculty of Basic Sciences

Week_06

Fall_2026

Weekly Plan



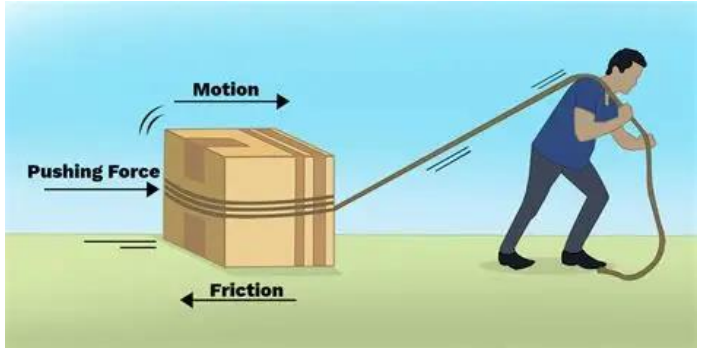
- **Applying Newton’s Law**
- **Work and Kinetic Energy**

Forces (Contact Forces)

Forces that occur when two objects are physically touching each other.

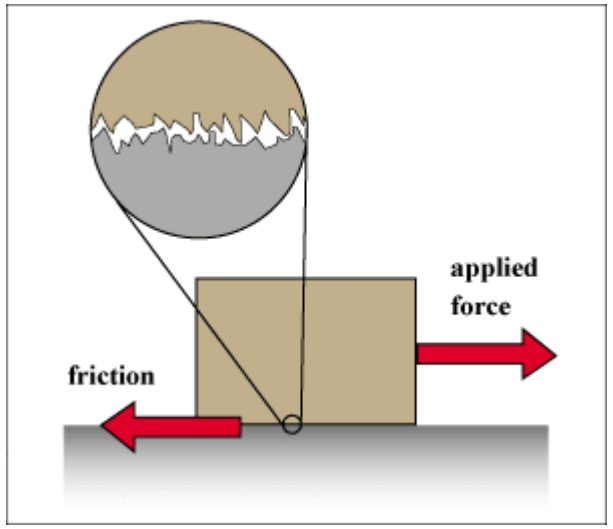
Force Type	Description	Real-World Example
Frictional Force	Opposes relative motion between surfaces in contact	Walking, car brakes, sliding a book
Normal Force	Perpendicular force exerted by a surface on an object	Book resting on a table, person standing on floor
Tension Force	Pulling force through a string, rope, or cable	Towing a car, elevator cable, hanging sign
Applied Force	Direct push or pull on an object	Pushing a shopping cart, kicking a ball
Air Resistance	Frictional force from air molecules on moving objects	Parachute, skydiver falling
Spring Force	Restoring force from compressed/stretched spring	Car suspension, mattress springs
Buoyant Force	Upward force from fluid on submerged objects	Boat floating, helium balloon rising

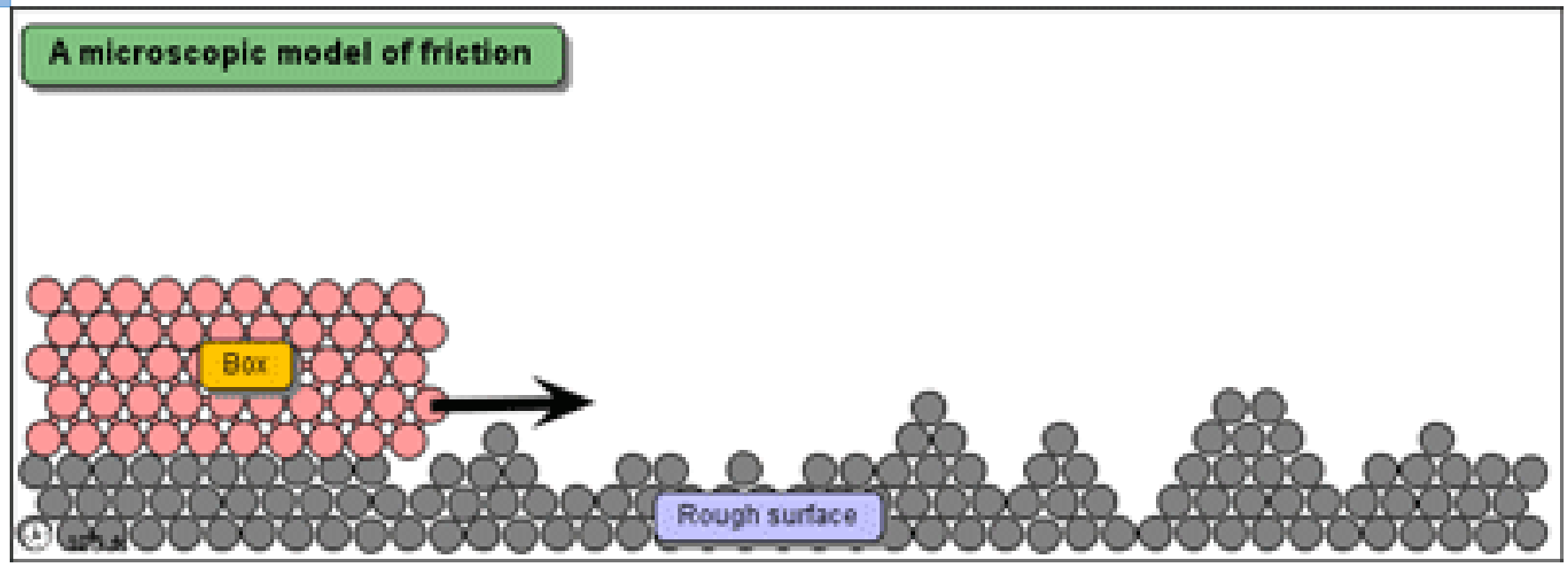
Friction



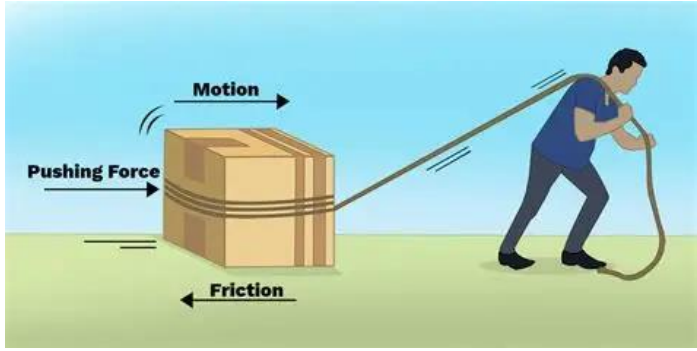
$$f_k \propto n$$

$$f_k = \mu_k n$$



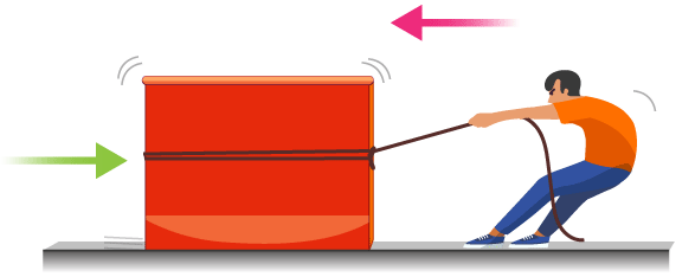


Friction

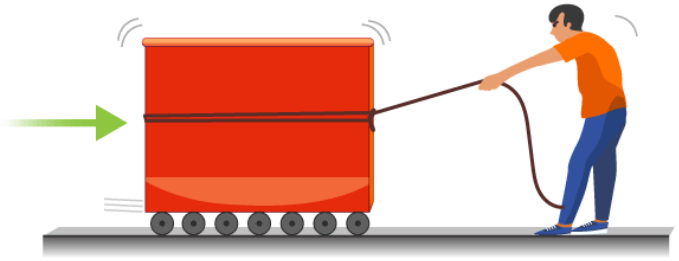


$$f_k \propto n$$

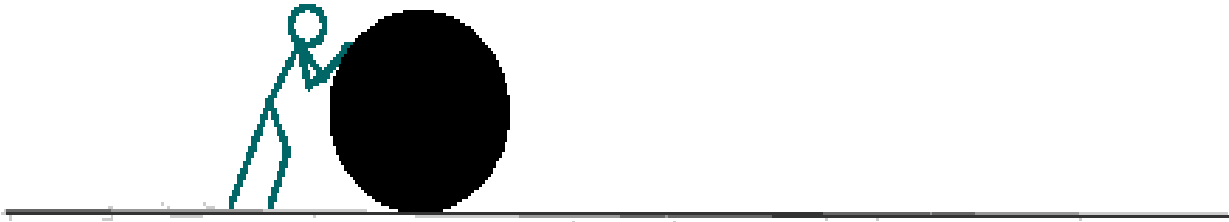
$$f_k = \mu_k n$$



Sliding Friction



Rolling Friction





Friction

$$f_k = \mu_k n$$

$$f_S = \mu_S n$$

System	Static Friction μ_s	Kinetic Friction μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.5-0.7	0.3-0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Metal on wood	0.5	0.3
Steel on steel (dry)	0.6	0.3
Steel on steel (oiled)	0.05	0.03
Teflon on steel	0.04	0.04
Bone lubricated by synovial fluid	0.016	0.015
Shoes on wood	0.9	0.7
Shoes on ice	0.1	0.05
Ice on ice	0.1	0.03
Steel on ice	0.4	0.02

Magnitude of Static Friction

The magnitude of static friction f_s is

$$f_s \leq \mu_s N,$$

where μ_s is the coefficient of static friction and N is the magnitude of the normal force.

Magnitude of Kinetic Friction

The magnitude of kinetic friction f_k is given by

$$f_k = \mu_k N,$$

where μ_k is the coefficient of kinetic friction.

Example

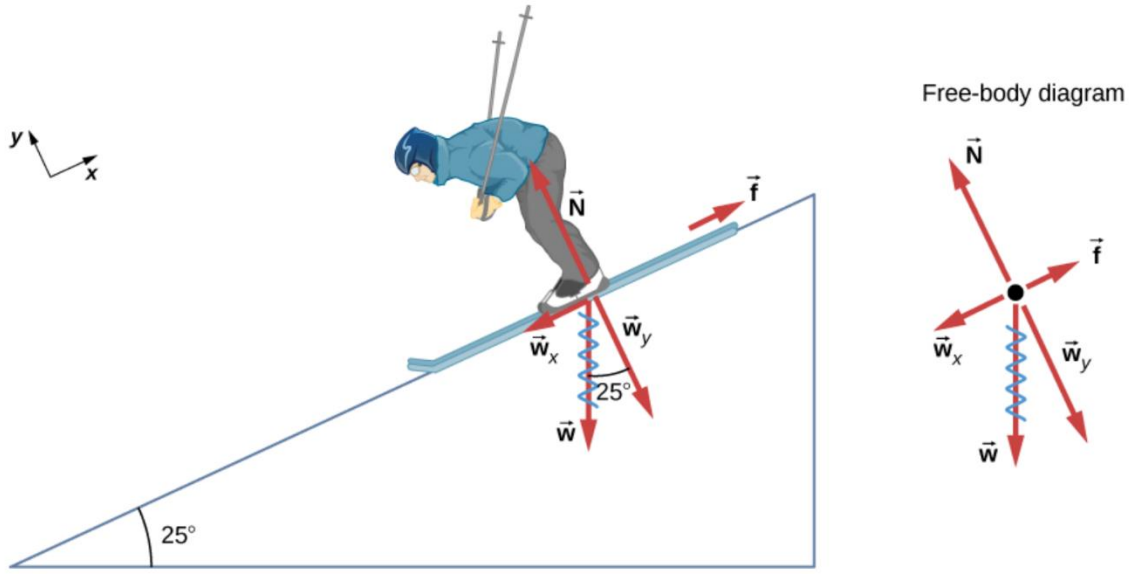
The skier on the slope in Figure. His mass including equipment is 60.0 kg. (a) What is his acceleration if friction is negligible? (b) What is his acceleration if friction is 45.0 N?

$$w_x = w \sin 25° = mg \sin 25°,$$

$$w_y = w \cos 25° = mg \cos 25°.$$

$$F_{\text{net}\,x} = w_x - f.$$

$$a_x = \frac{F_{\text{net}\,x}}{m}$$



Example

The skier on the slope in Figure. His mass including equipment is 60.0 kg. (a) What is his acceleration if friction is negligible? (b) What is his acceleration if friction is 45.0 N?

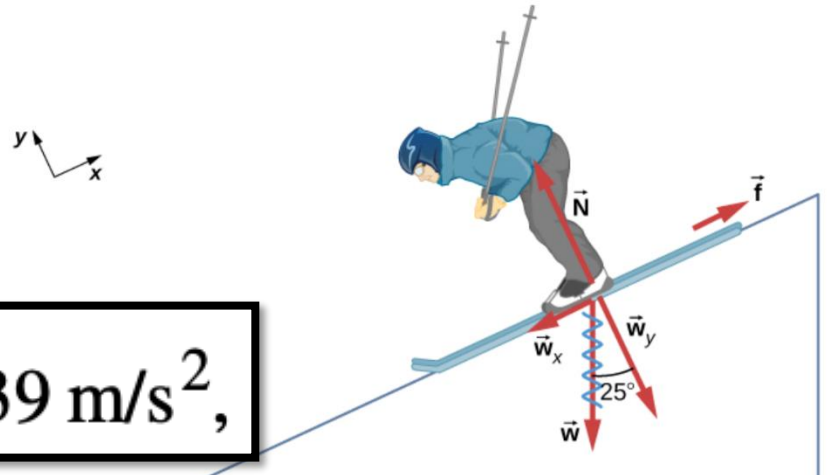
$$F_{\text{net}\,x} = w_x - f.$$

$$a_x = \frac{F_{\text{net}\,x}}{m}$$

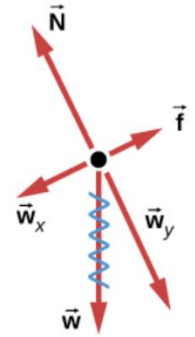
$$a_x = \frac{F_{\text{net}\,x}}{m} = \frac{w_x - f}{m} = \frac{mg\sin 25° - f}{m}.$$

$$a_x = \frac{(60.0\ \text{kg})(9.80\ \text{m/s}^2)(0.4226) - 45.0\ \text{N}}{60.0\ \text{kg}}.$$

$$a_x = 3.39\ \text{m/s}^2,$$



Free-body diagram



$$a_x = \frac{F_{\text{net}\,x}}{m} = \frac{mg\sin 25°}{m} = g\sin 25°$$

$$(9.80\ \text{m/s}^2)(0.4226) = 4.14\ \text{m/s}^2$$

Friction

$$f_k = \mu_k n$$

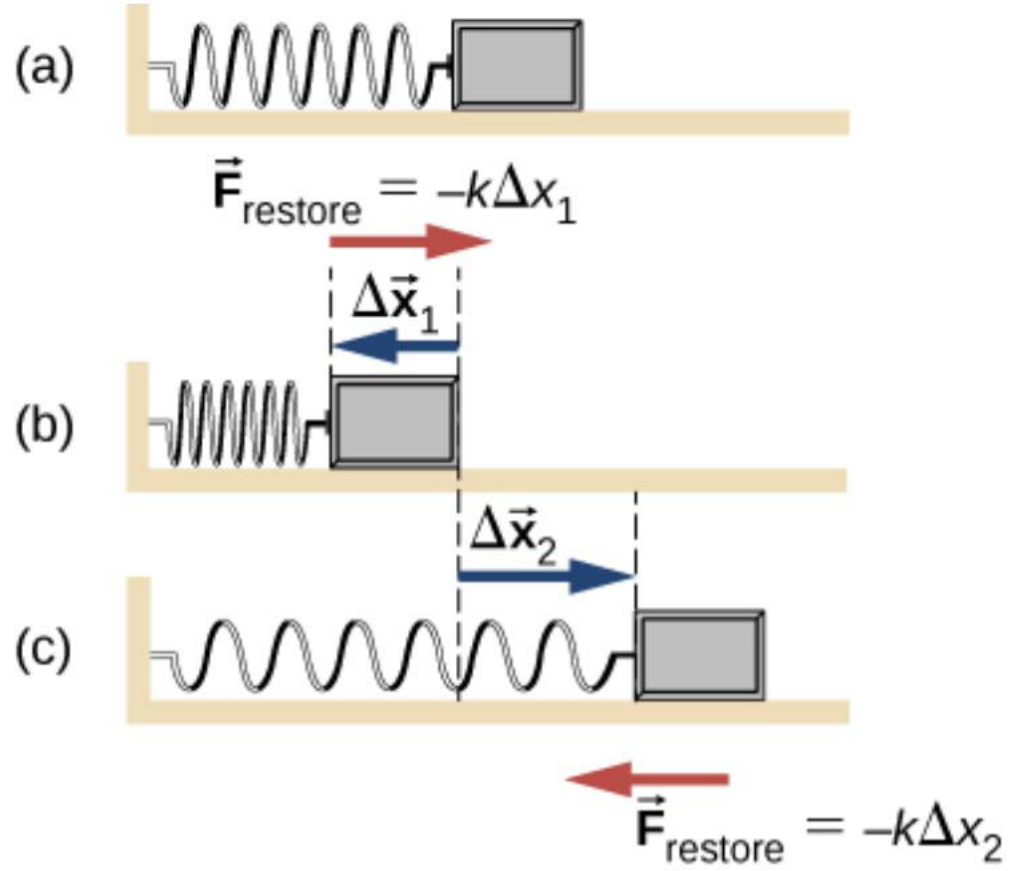
$$f_s = \mu_s n$$

$$\vec{\mathbf{F}} = -k\,\vec{\mathbf{x}}$$

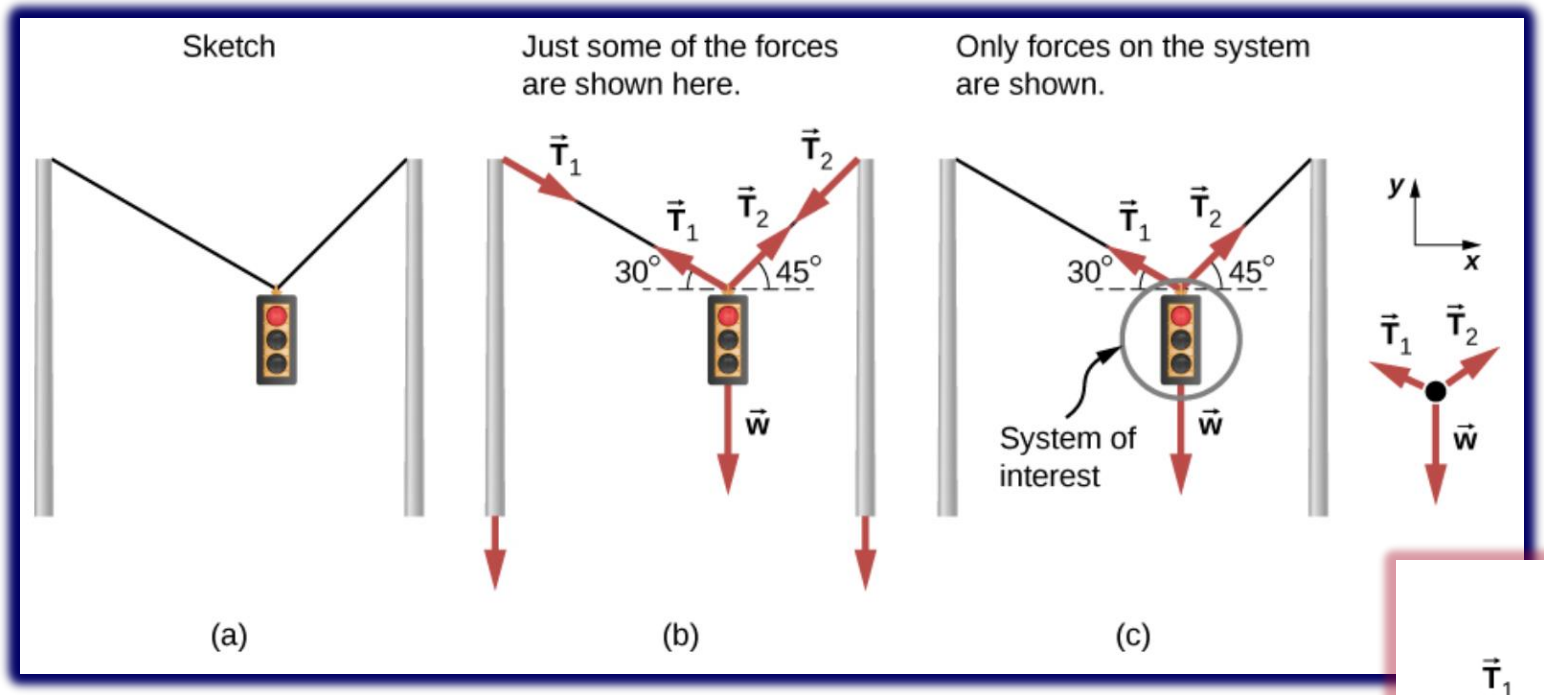
(a)The spring is in a relaxed position and exerts no force on the block.

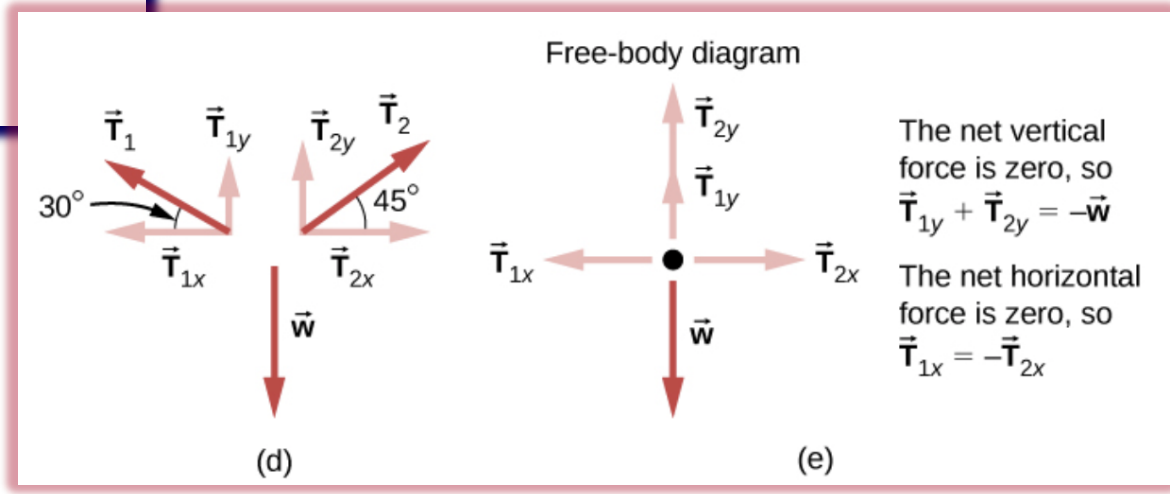
(b)The spring is compressed by displacement $\Delta\vec{\mathbf{x}}_1$ of the object and exerts restoring force $-k\Delta\vec{\mathbf{x}}_1$.

(c)The spring is stretched by displacement $\Delta\vec{\mathbf{x}}_2$ of the object and exerts restoring force $-k\Delta\vec{\mathbf{x}}_2$.

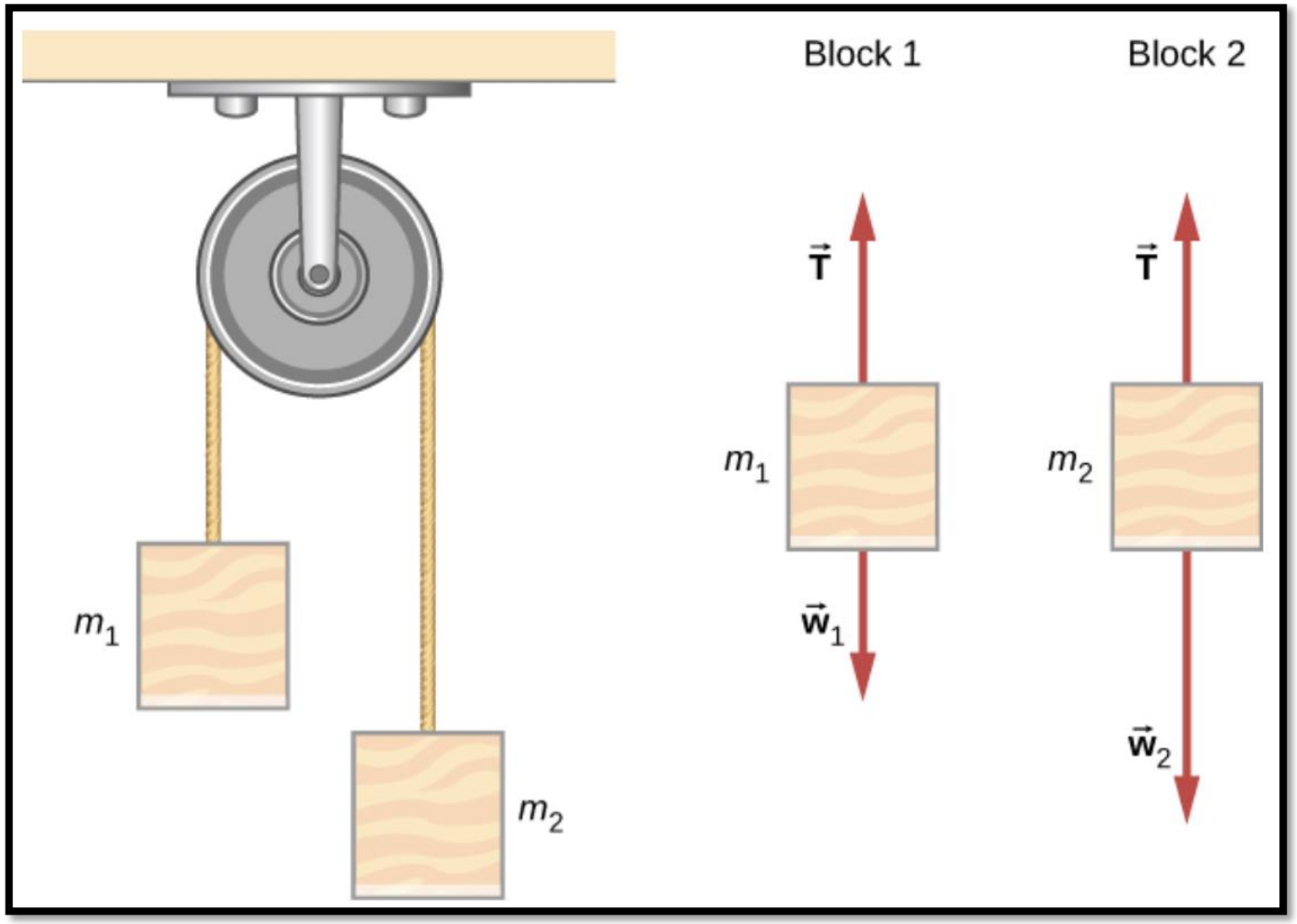


Forces


Sketch
Just some of the forces are shown here.
Only forces on the system are shown.
$\vec{T}_1$
$\vec{T}_2$
30°
45°
$\vec{w}$
System of interest
y
x
(a)
(b)
(c)


Free-body diagram
$\vec{T}_1$
$\vec{T}_{1y}$
$\vec{T}_{2y}$
$\vec{T}_2$
30°
45°
$\vec{T}_{1x}$
$\vec{T}_{2x}$
$\vec{w}$
The net vertical force is zero, so $\vec{T}_{1y} + \vec{T}_{2y} = -\vec{w}$
The net horizontal force is zero, so $\vec{T}_{1x} = -\vec{T}_{2x}$
(d)
(e)

Forces

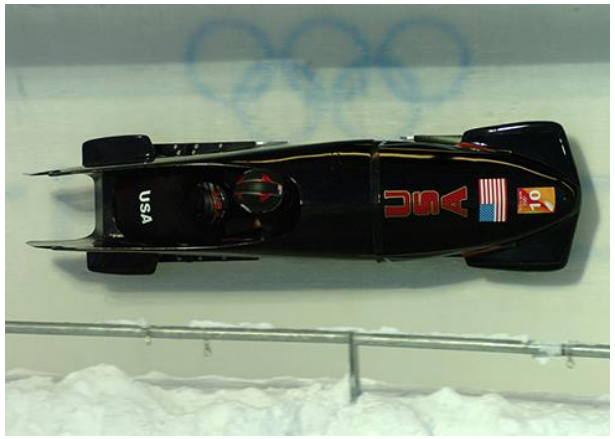


Forces (Drag and Terminal Speed)

$$F_D \propto v^2.$$

Drag force acts on an object when it is moving in a fluid (either a gas or a liquid). The drag force always opposes the motion of an object.

$$F_D = \frac{1}{2} C \rho A v^2,$$





C is the drag coefficient,
A is the area of the object facing the fluid,
ρ is the density of the fluid.
This equation can also be written in a more generalized fashion as $F_D = bv^2$, where *b* is a constant equivalent to $0.5C\rho A$.

Object	*C*	Object	*C*
Airfoil	0.05	Sphere	0.45
Toyota Camry	0.28	Hummer H2 SUV	0.64
Ford Focus	0.32	Skydiver (feet first)	0.70
Honda Civic	0.36	Bicycle	0.90
Ferrari Testarossa	0.37	Skydiver (horizontal)	1.0
Dodge Ram Pickup	0.43	Circular flat plate	1.12

Work

W=?

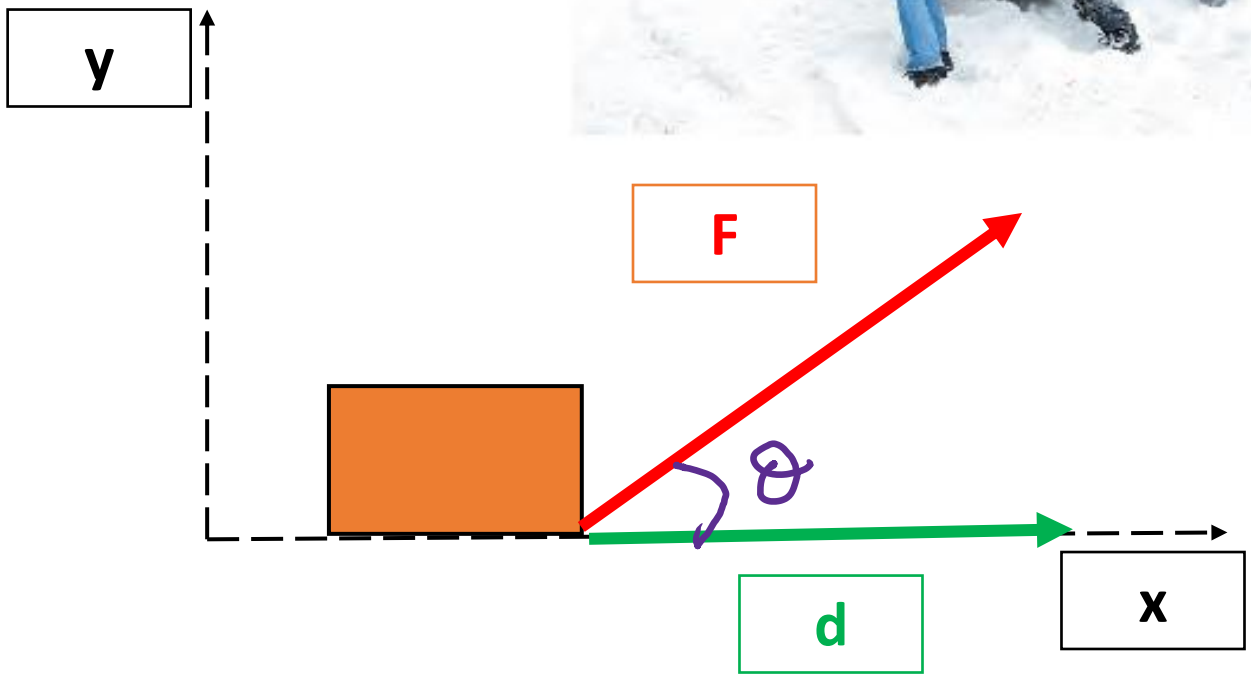
$$W = \boldsymbol{F}.\boldsymbol{d}$$

$$W = F_x d$$

$$W = Fd\,Cos(\Theta)$$

Figure 6.1 These people are doing work as they push on the car because they exert a force on the car as it moves.





Work

$\theta = 0$

$W = Fd$

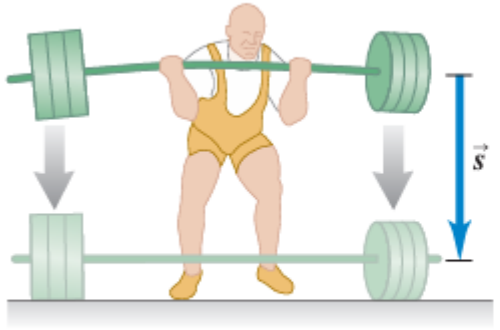
$\theta = 90$

$W = 0$

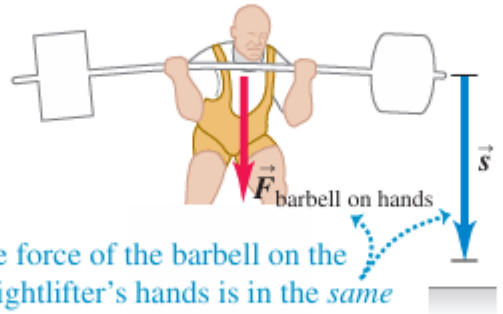
$\theta = 180$

$W = -Fd$

(a) A weightlifter lowers a barbell to the floor.

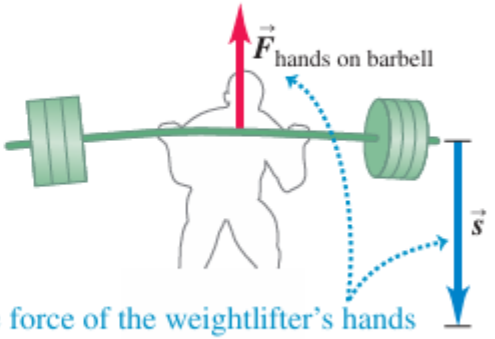


(b) The barbell does *positive* work on the weightlifter's hands.



The force of the barbell on the weightlifter's hands is in the *same* direction as the hands' displacement.

(c) The weightlifter's hands do *negative* work on the barbell.



The force of the weightlifter's hands on the barbell is *opposite* to the barbell's displacement.

d

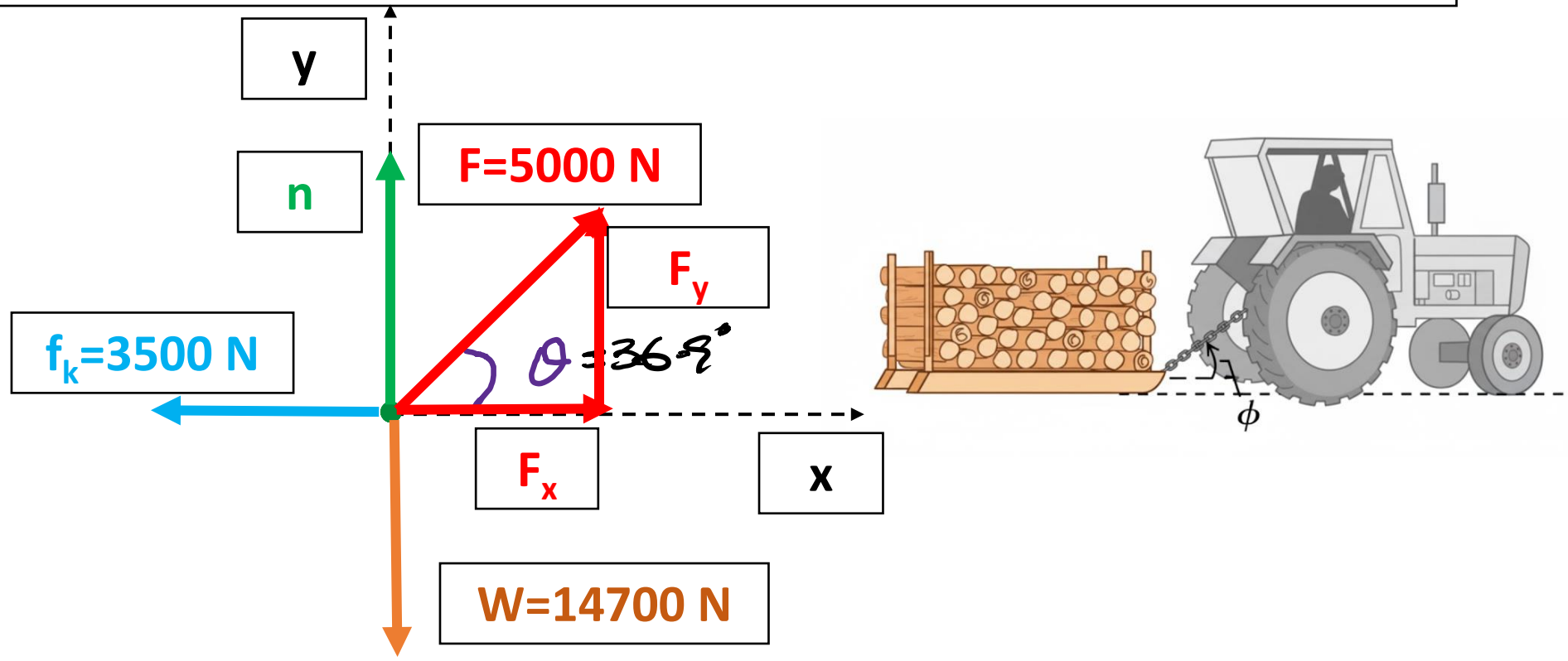
x

Condition	Work	Example
$\cos\theta > 0$	Positive	Pushing a car forward
$\cos\theta < 0$	Negative	Braking a moving car
$\cos\theta = 0$	Zero	Carrying a load horizontally

Example

A farmer hitches his tractor to a sled loaded with firewood and pulls it a **distance of 20 m** along level ground. The total **weight** of sled and load is **14,700 N**. The tractor exerts a constant **5000-N force** at an **angle of 36.9°** above the horizontal. **A 3500-N friction force** opposes the sled's motion. **Find the work done by each force** acting on the sled and the total work done by all the forces.

W=?



Example

A farmer pull his tractor to a sled loaded with firewood and pulls it a **distance of 20 m** along level ground. The total **weight** of sled and load is **14,700 N**. The tractor exerts a constant **5000-N force** at an **angle of 36.9°** above the horizontal. **A 3500-N friction force** opposes the sled's motion. **Find the work done by each force** acting on the sled and the total work done by all the forces.

$$W = (F_{net})(d)$$

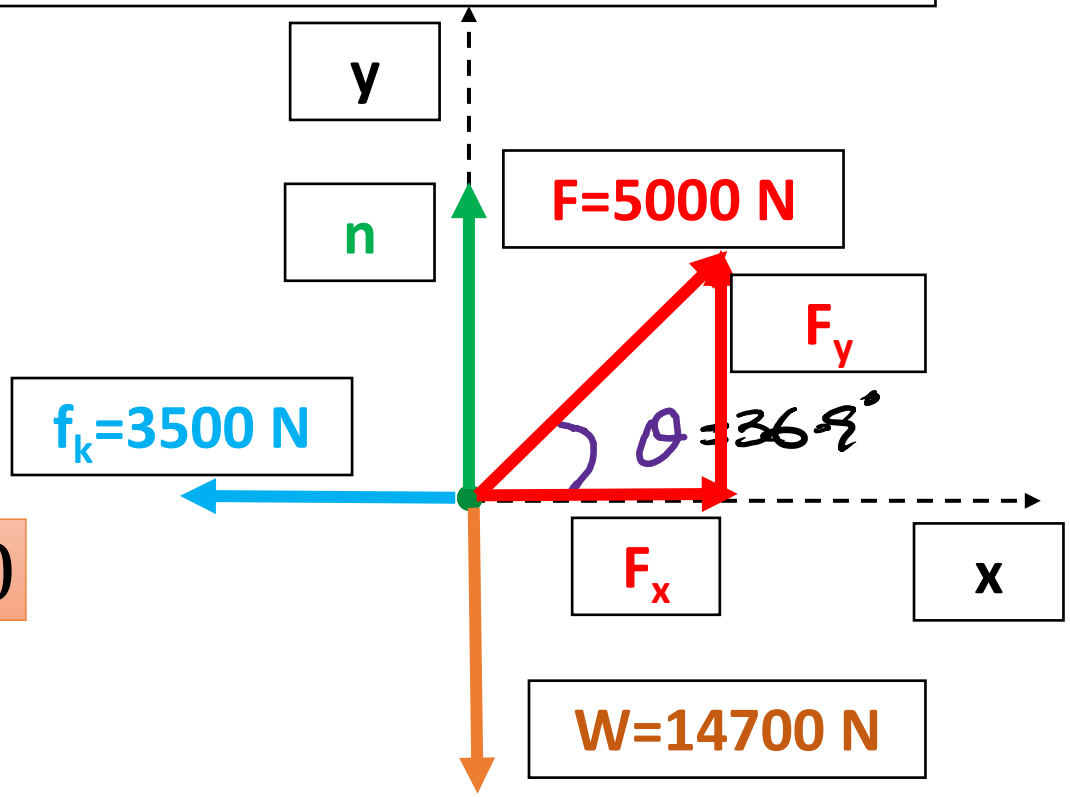
W=?

$$W = (F_{net})(20)$$

$$F_{net} = F_x - f_k$$

$$F_{net} = 5000\cos(36.9) - 3500 = 500$$

$$W = (500)(20)\ J$$



Work-Energy Theorem

Net work done on a body is equal to the change in its kinetic energy

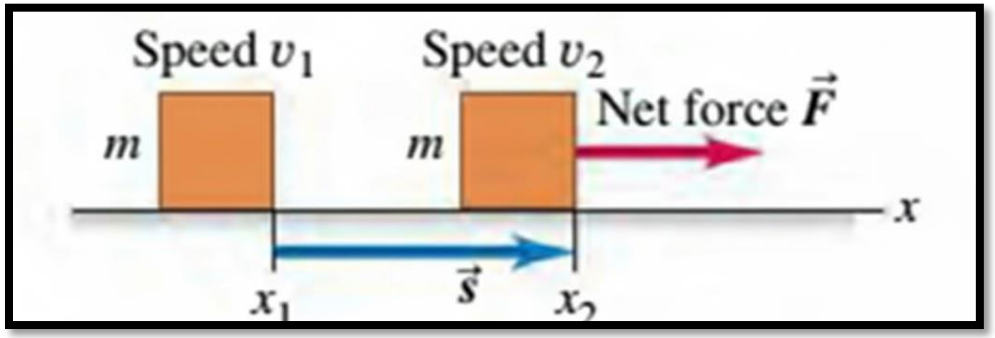
$$V_2^2 = V_1^2 + 2a_xS$$

$$V_2^2 - V_1^2 = 2a_xS$$

$$(V_2^2 - V_1^2)/2s = a_x$$

$$ma_x = m(V_2^2 - V_1^2)/2s$$

$$F = m(V_2^2 - V_1^2)/2s$$



$$Fs = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$$

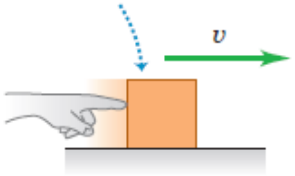
$$W = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$$

$$W = (KE)_f - (KE)_i$$

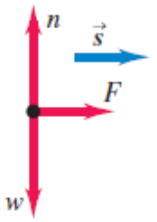
Figure 6.8 The relationship between the total work done on an object and how the object's speed changes.

(a)

A block slides to the right on a frictionless surface.

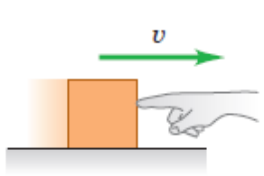


If you push to the right on the block as it moves, the net force on the block is to the right.

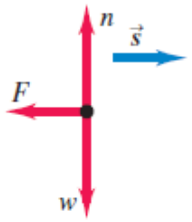


- The total work done on the block during a displacement $\vec{s}$ is positive: $W_{tot} > 0$.
- The block speeds up.

(b)

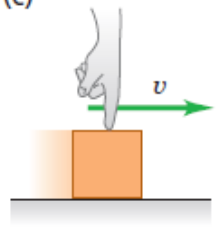


If you push to the left on the block as it moves, the net force on the block is to the left.



- The total work done on the block during a displacement $\vec{s}$ is negative: $W_{tot} < 0$.
- The block slows down.

(c)

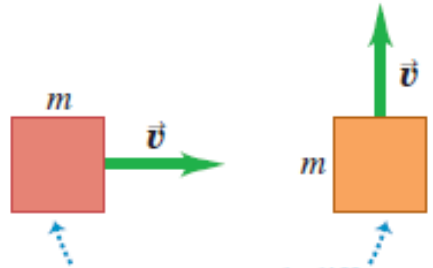


If you push straight down on the block as it moves, the net force on the block is zero.

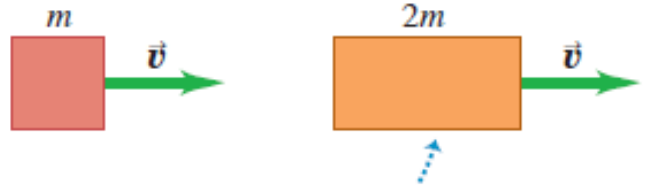


- The total work done on the block during a displacement $\vec{s}$ is zero: $W_{tot} = 0$.
- The block's speed stays the same.

Figure 6.10 Comparing the kinetic energy $K = \frac{1}{2}mv^2$ of different objects.



Same mass, same speed, different directions of motion: *same* kinetic energy



Twice the mass, same speed: *twice* the kinetic energy



Same mass, twice the speed: *four times* the kinetic energy

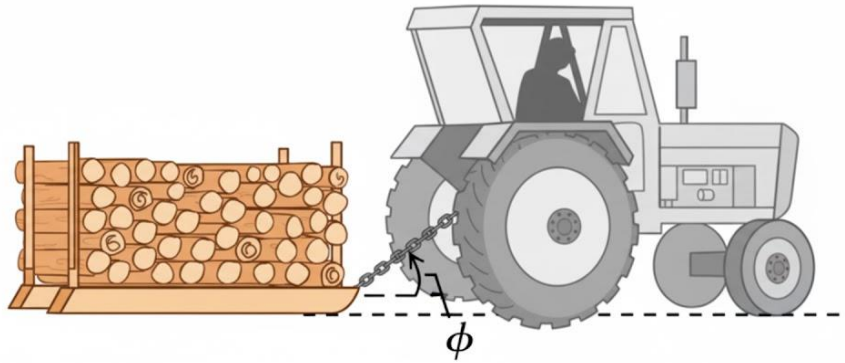
Example

A farmer hitches her tractor to a sled loaded with firewood and pulls it a **distance of 20 m** along level ground. The total **weight** of sled and load is **14,700 N**. The tractor exerts a constant **5000-N force** at an **angle of 36.9°** above the horizontal. **A 3500-N friction force** opposes the sled's motion. Suppose the sled's initial speed v_1 **is 2.0** m/s. **What is the speed** of the sled after **it moves 20 m**?

V_2=?

$$W = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$$

$$F_{net}d = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$$



$$F_{net} = F_x - f_k$$

$$F_{net} = 5000\cos(36.9) - 3500 = 500\ N$$

Example

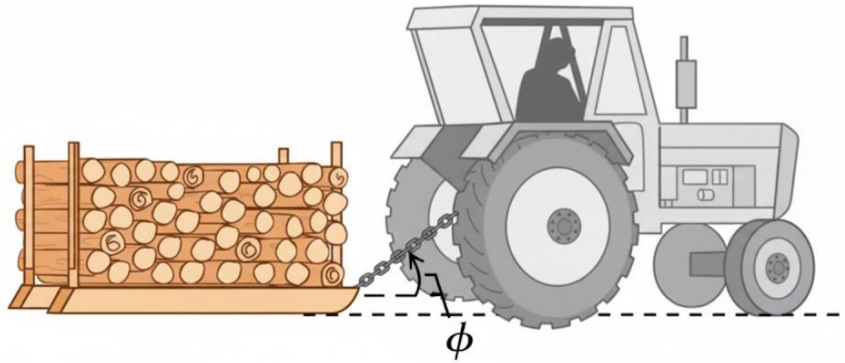
A farmer hitches her tractor to a sled loaded with firewood and pulls it a **distance of 20 m** along level ground. The total **weight** of sled and load is **14,700 N**. The tractor exerts a constant **5000-N force** at an **angle of 36.9°** above the horizontal. **A 3500-N friction force** opposes the sled's motion. Suppose the sled's initial speed v_1 **is 2.0** m/s. **What is the speed** of the sled after **it moves 20 m**?

V_2=?

$$(500)(20) = \frac{1}{2}(1500)V_2^2 - \frac{1}{2}(1500)(2)^2$$

$$V_2 = ?$$

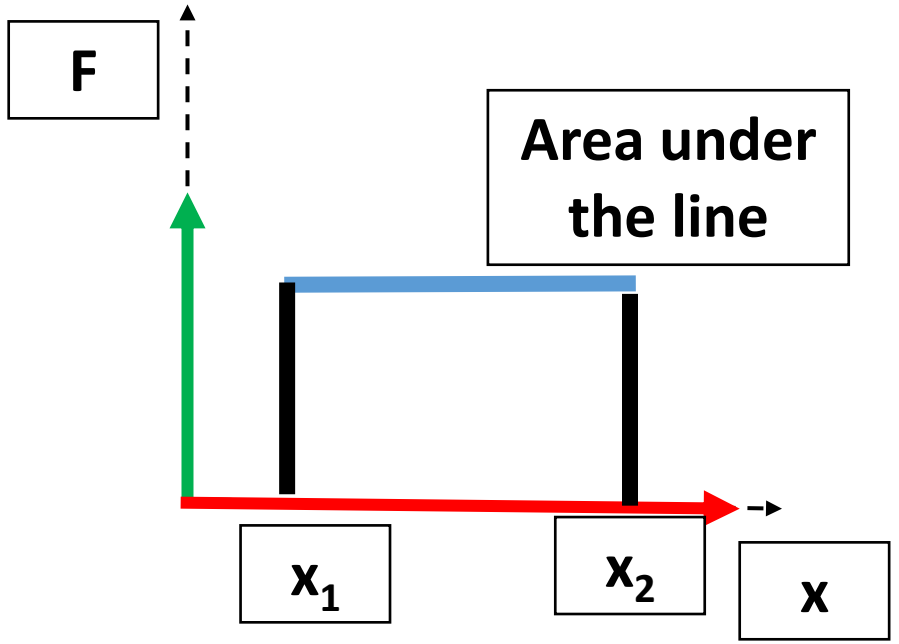
$$V_2 \approx \pm 4.16 \text{m/s}$$



Work

Constant Force

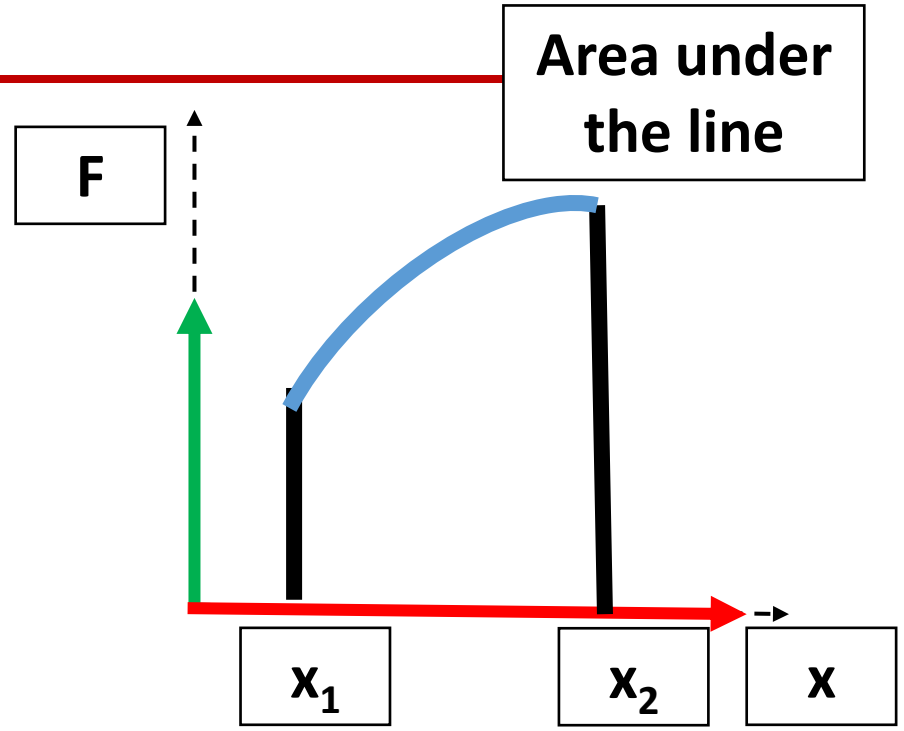
$$W = F(x_2 - x_1)$$



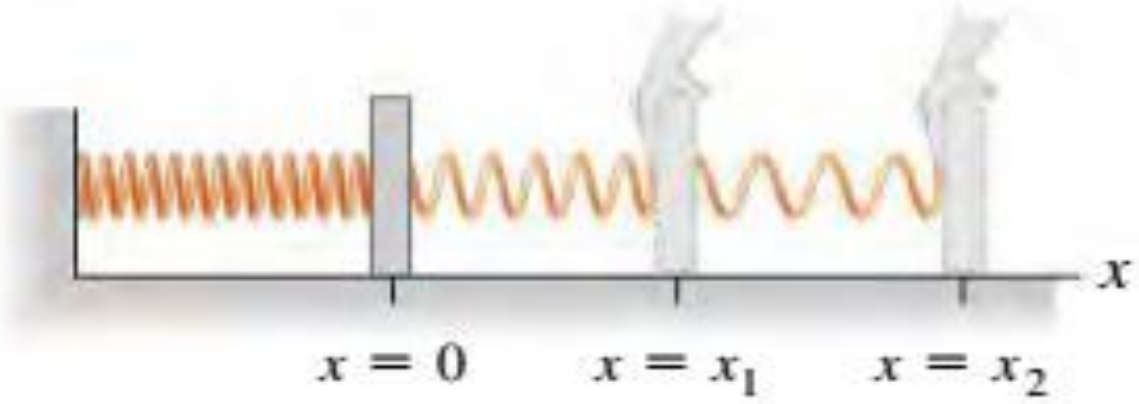
Work with Variable Force

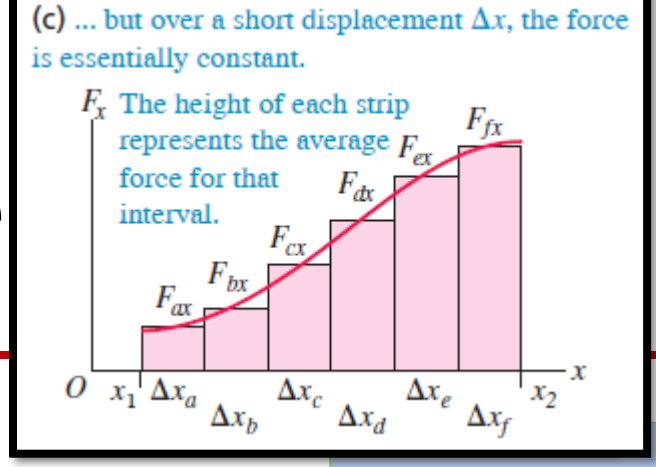
Variable Force

$$W = \int_{x_1}^{x_2} F(x)dx$$



Work with Variable Force



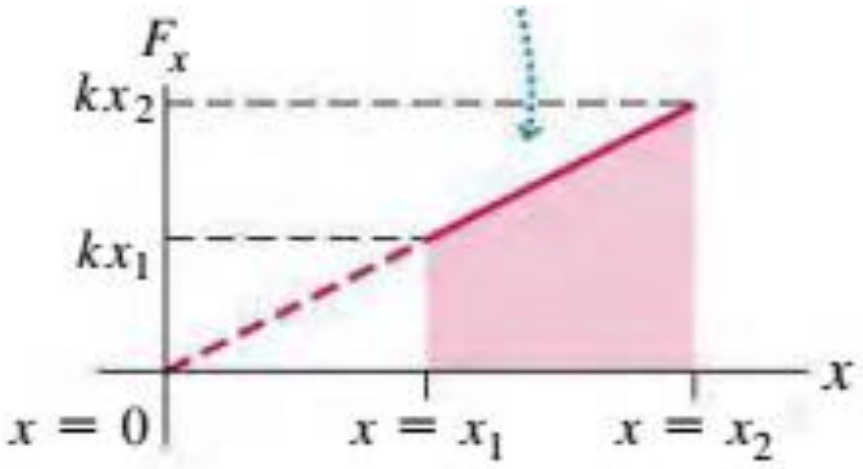


$$W = K\int_{x_1}^{x_2} x\,dx$$

$$W = \frac{1}{2}Kx_2^2 - \frac{1}{2}Kx_1^2$$

$$W = \int_{x_1}^{x_2} F(x)\,dx$$

$$W = \int_{x_1}^{x_2} (Kx)\,dx$$



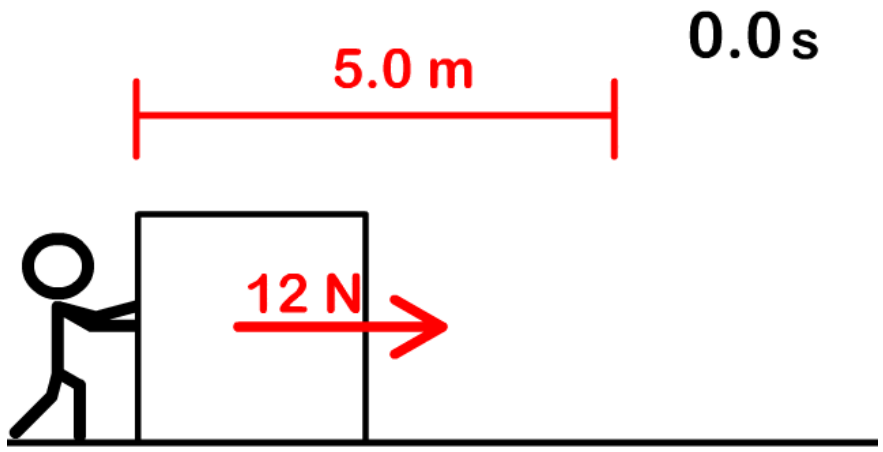
Power

$$P = \frac{W}{t}$$

$$P = \frac{\Delta W}{\Delta t}$$

$$P = \Delta t \to 0 \frac{\Delta W}{\Delta t}$$

$$P = \frac{dW}{dt}$$



Power

$$P = \frac{W}{t}$$

$$P = \frac{\Delta W}{\Delta t}$$

$$P = \lim_{t \to 0} \frac{\Delta W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

$$Watt = \frac{J}{s}$$

$$P = \frac{F \Delta s}{\Delta t}$$

$$P = F \lim_{t \to 0} \frac{\Delta s}{\Delta t}$$

$$P = F\, V$$

$$P = \boldsymbol{F}.\boldsymbol{V}$$

$$P = KW$$

$$E = Pt = KWh$$

Power

$$Watt = \frac{J}{s}$$

$$P =? \; W = hp$$

$$E = Pt = KWh$$

$$E = Pt = KWh$$
$$= 1000 * \left(\frac{J}{s}\right) * (3600s)$$

$$E = Pt = KWh = 3.6 \times 10^6 J$$

Example

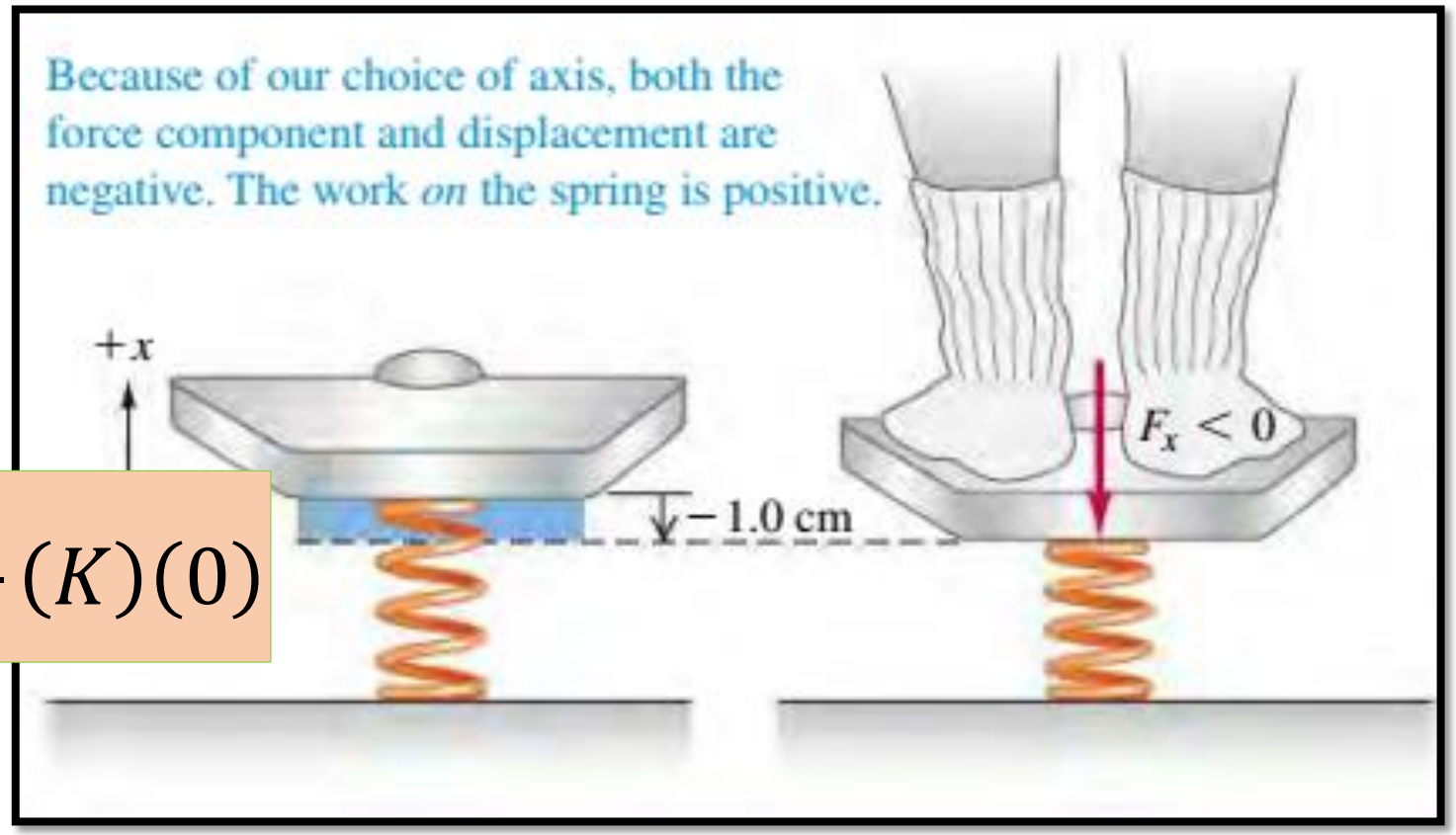
A person **weighing 600 N** steps on a bathroom scale that contains a stiff spring. In equilibrium, the **spring is compressed 1.0 cm** under her weight. **Find the force constant of the spring** and the total work done on it during the compression.

$$W = \frac{1}{2}Kx_2^2 - \frac{1}{2}Kx_1^2$$

$$K = \frac{F_x}{x} = \frac{-600}{-0.01} = 6X10^4 N/m$$

$$W = \frac{1}{2}(6X104)(-0.01)^2 - \frac{1}{2}(K)(0)$$

$$W = 3\ J$$



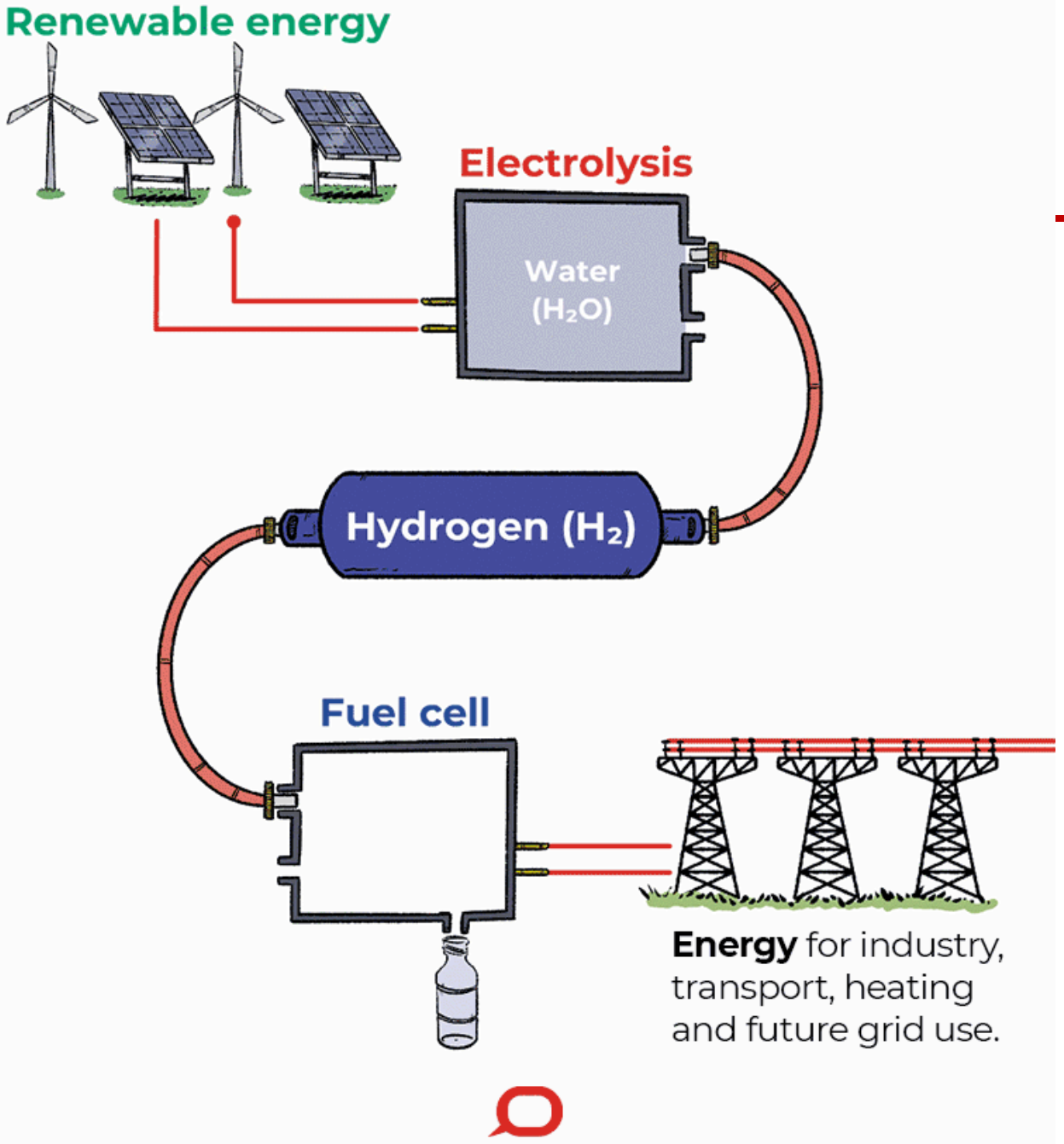
Example

5.3, 5.6, 5.15, 5.24, 5.33, 5.51, 5.96
6.7, 6.8, 6.25, 6.34, 6.46, 6.56



What is the speed of arrow?

How sustainable hydrogen energy works:



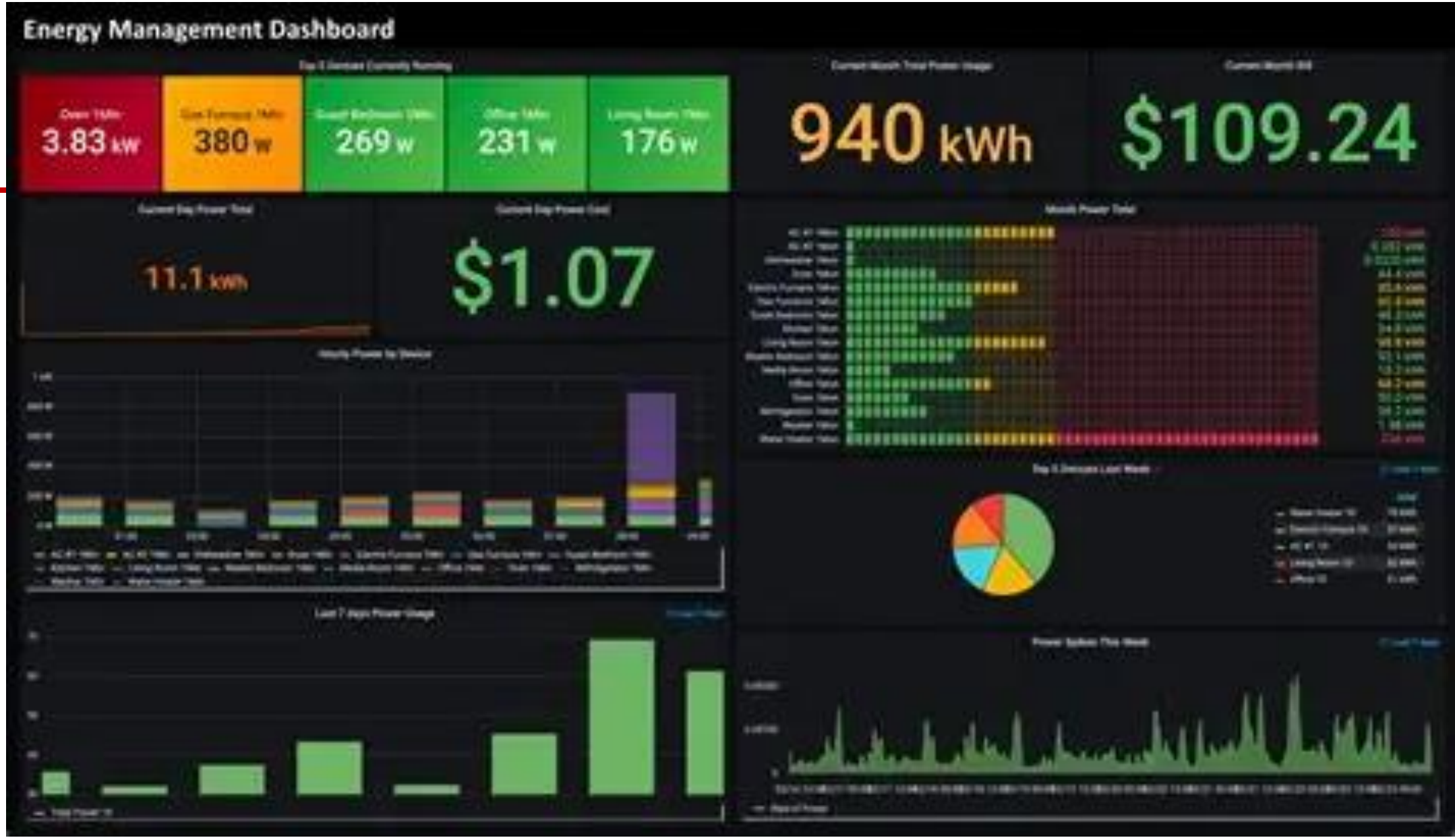
Energy


POTENTIAL ENERGY
(Stored Energy)
GRAIVATIONAL
CHEMICAL
NUCLEAR
ELASTIC
Energy due due height or position.
Example: Wrecking ball, water behind dam.
Energy in molecular bonds.
Example:
Food, Batteries, Fuels.
Energy in atomic nucleus.
Example: Power
Example; Plants, Sun
Energy from tension/compression.
Example: Stretthed rubber rabard, Compressed spring.
Stored energy with the potential to do work.

Energy


KINETIC ENERGY
(Energy of Motion)
MOTION
THERMAL
ELECTRICAL
SOUND
Energy of moving objects.
Example: Car, wind, water.
THERMAL (Heat)
Energy from particle.
Example: Fire, stove, agothemal.
Energy from particle motion.
Example: Fire, stove, appliances.
RADIANT (Light)
Energy from vibrations.
Speaker, clapping.
Energy due to moveement.

EMS


Energy Management Dashboard
3.83 kW
380 W
269 W
231 W
176 W
940 kWh
$109.24
11.1 kWh
$1.07

Energy & Emissions

Energy Usage

TJP 6.53 MW

TGG 7.22 MW

Real Time Data | 7 Day Average

Description	Min	Max
TWP1 EC	0	15
TWP2 EC	0	8
WWPS EC	0	16
WWPT EC	0	18

Description	Units	Min	Ma
TWP1 EC	W/kW	0	15
TWP2 EC	W/kW	0	8
WWPS EC	W/kW	0	16
WWPT EC	W/kW	0	18

15.0 / 12.0 / 9.0 / 6.0 / 3.0 / 0.0

4/25 4/27 4/29

Tag	Description
PM1.EnergyCost	Energy Cost - Electric
PM2.EnergyCost	Energy Cost - Total
Total.EnergyCost	Energy Cost - Gas

200.0 / 160.0 / 120.0 / 80.0 / 40.0 / 0.0

5/1/24 5/1/24

Excel Sheets | South Lab | Env. Lab | Reports

Water Usage

Process Area Usage	Total Usage
5148233	Steam, Ect
28VFT8235	Feedwater

CO2 Emissions

Description	Units	Min	Max	Current
Total Emissions		0	25	4.9

Description	Min	Ma	Current
#2 Boiler CF	0	100	64.4

100.0 / 80.0 / 60.0 / 40.0 / 20.0 / 0.0

Description	Units	Current
#2 Boiler #2 CF	NULL	45.7

100.0 / 80.0 / 60.0 / 40.0 / 20.0 / 0.0

5/1/24

Online | Enviornmental



Centerline | More LAB Data | Waste Report

NH4	T1: 1.36 T2: 1.39 T3: 1.44	USL: 1.0 mg/L Target: 0.40 mg/L	pH	T: 7.1	USL: 7.6 Target: 7.0 USL: 6.8		
NO2	T1: 0.22 T2: 0.25 T3: 0.24	USL: 0.75 mg/L Target: 0.10 mg/L LSL: .05 mg/L	NO3	T1: 4.8 T2: 4.8 T3: 4.7	USL: 10.0 mg/L Target: 5.0 mg/L	TOC	T1: 48 T2: 47 T3: 44	USL: 50 mg/L Target: 20mg/L
Spot	T: 182	Target: 100 mg/L	Temp	T: 14	USL: 12 C Target: 8 C	TSS	T: 115	USL: 100 mg/L Target: 75 mg/L

Store and Retreat



Disposal to River / Lake



Energy Management Dashboard

HashStudioz Technologies

- Home
- Alarms
- Dashboards
- Entities
- Profiles
- Customers
- Rule chains
- Edge management
- Advanced features
- Resources
- Notification center

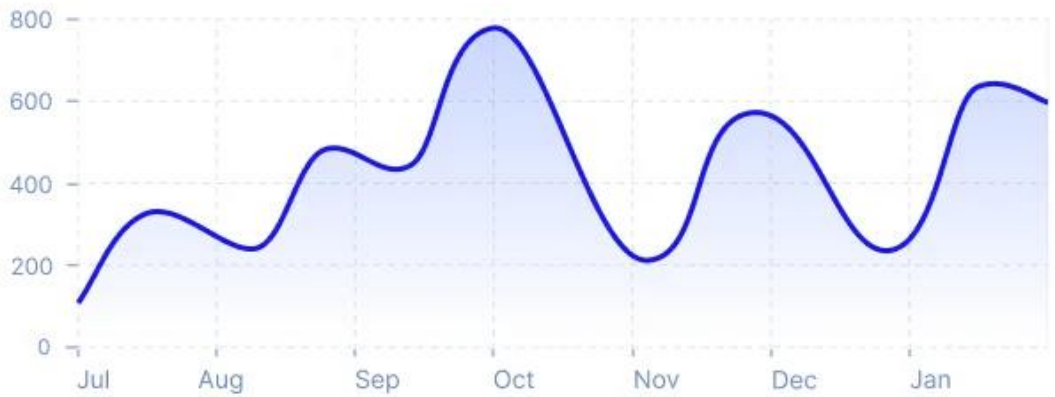
Dashboard

Dashboard

Khichdipur - Client Company

Client Name:	MCD Delhi
Location:	310, Golf Course Rd, IILM Institute, Sector 53, Gurugram
Model:	WTCJFJHB6896JJOKB
Date of Commissioning:	21-SEP-2021
Elefo Officer Incharge:	Mr. Siddharth Malhotra
Client Office Incharge:	Mr. Harshvardhan Rane

Unit consumed per month



Weekly Activity

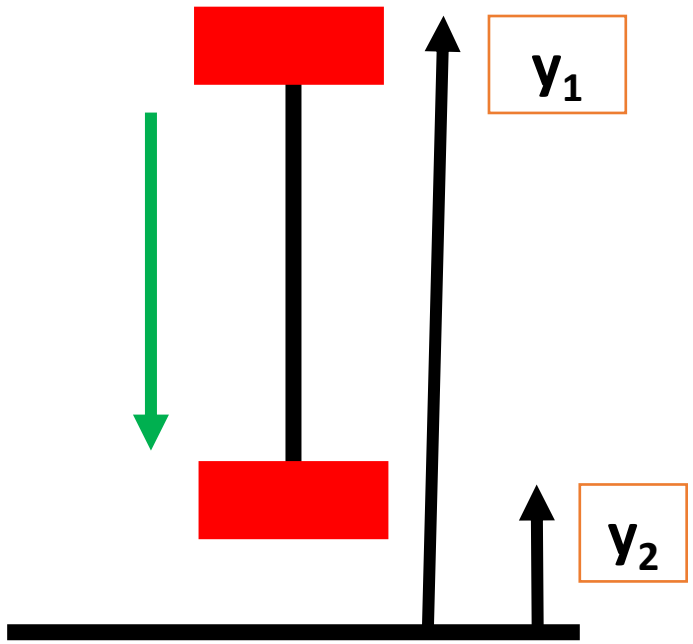


Energy Consumption



Summary

$$W + \Delta U = 0$$



$$W = (KE)_f - (KE)_i$$

$$W = -\Delta U$$

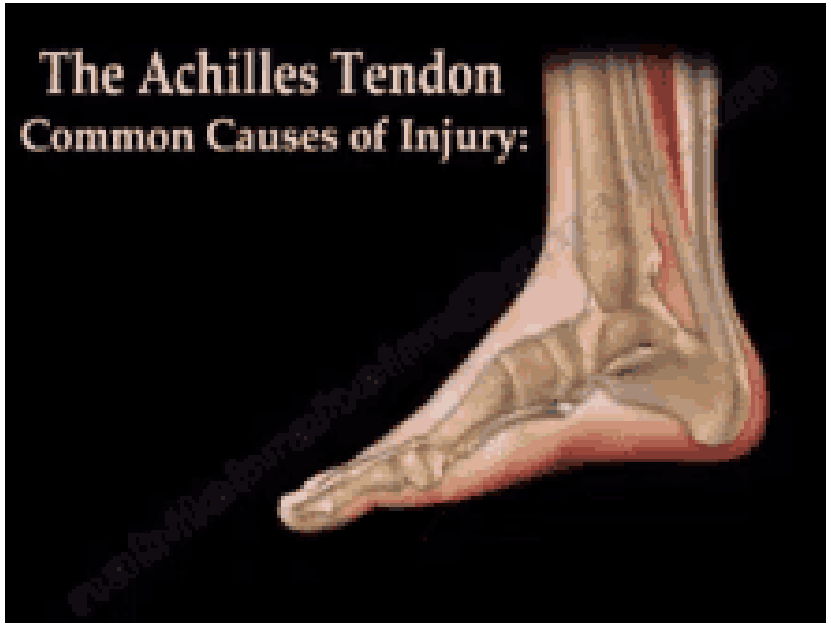
$$K_2 + U_2 = K_1 + U_1$$

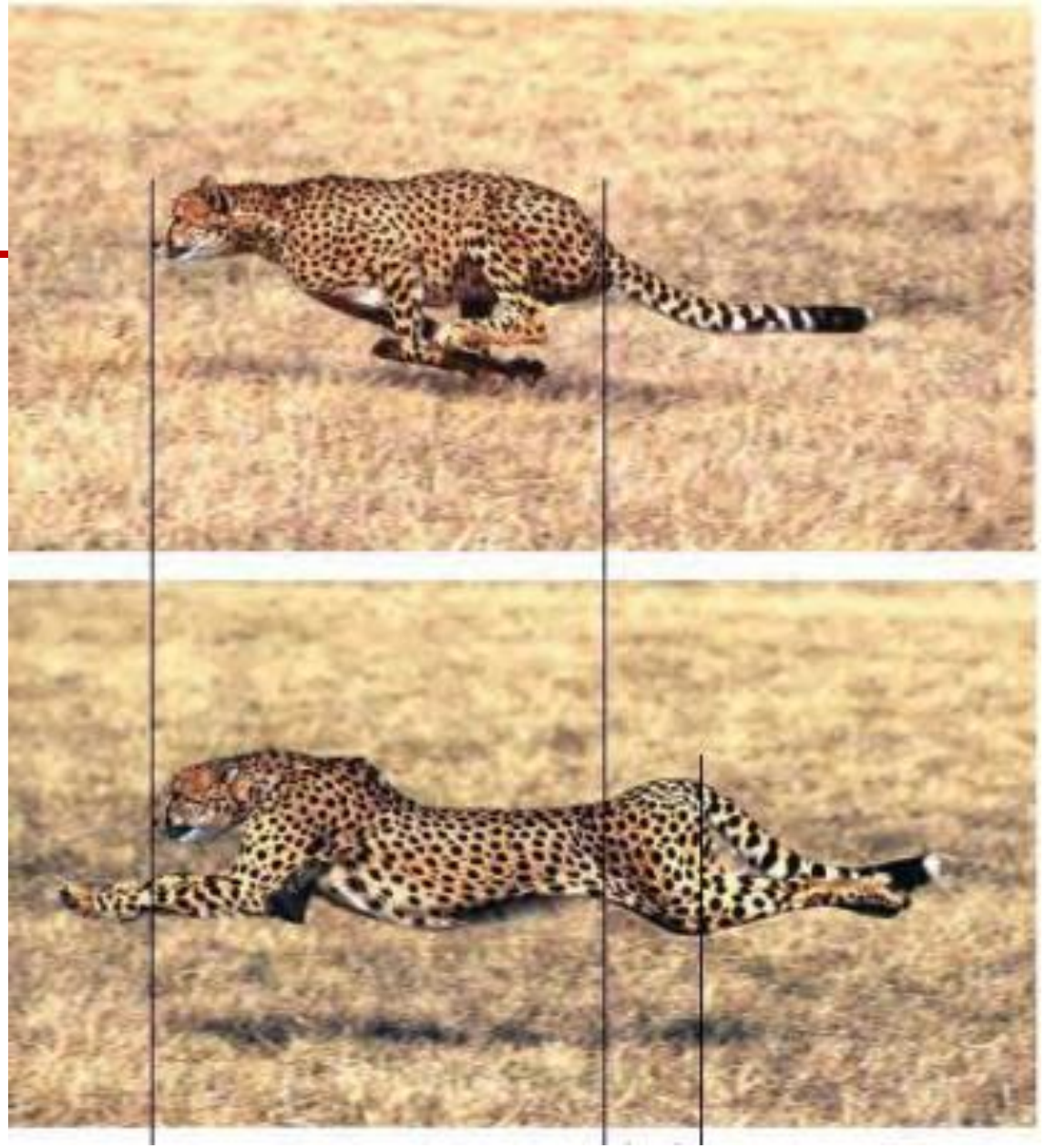
$$W_{other} + K_1 + U_1 = K_2 + U_2$$

$$U = mgh$$

Work against the friction/losses

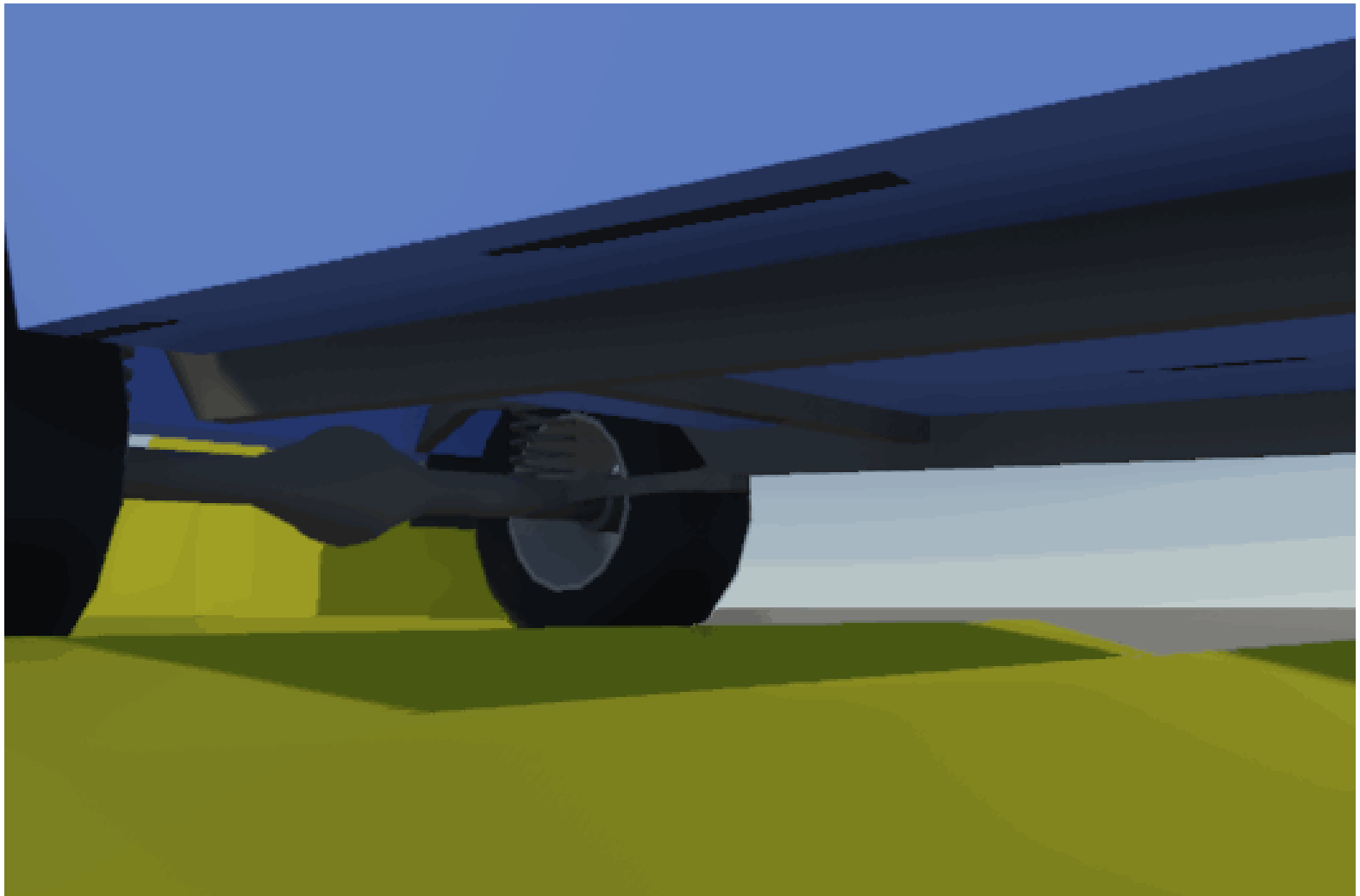
Elastic Potential Energy



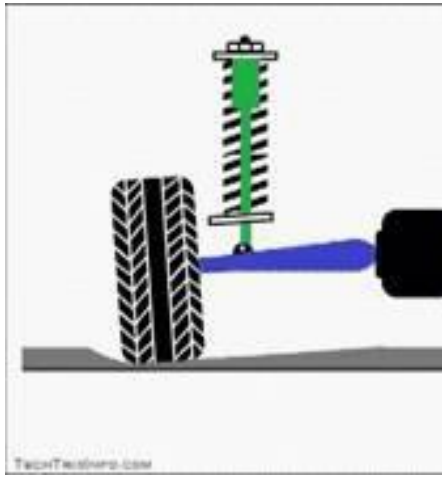


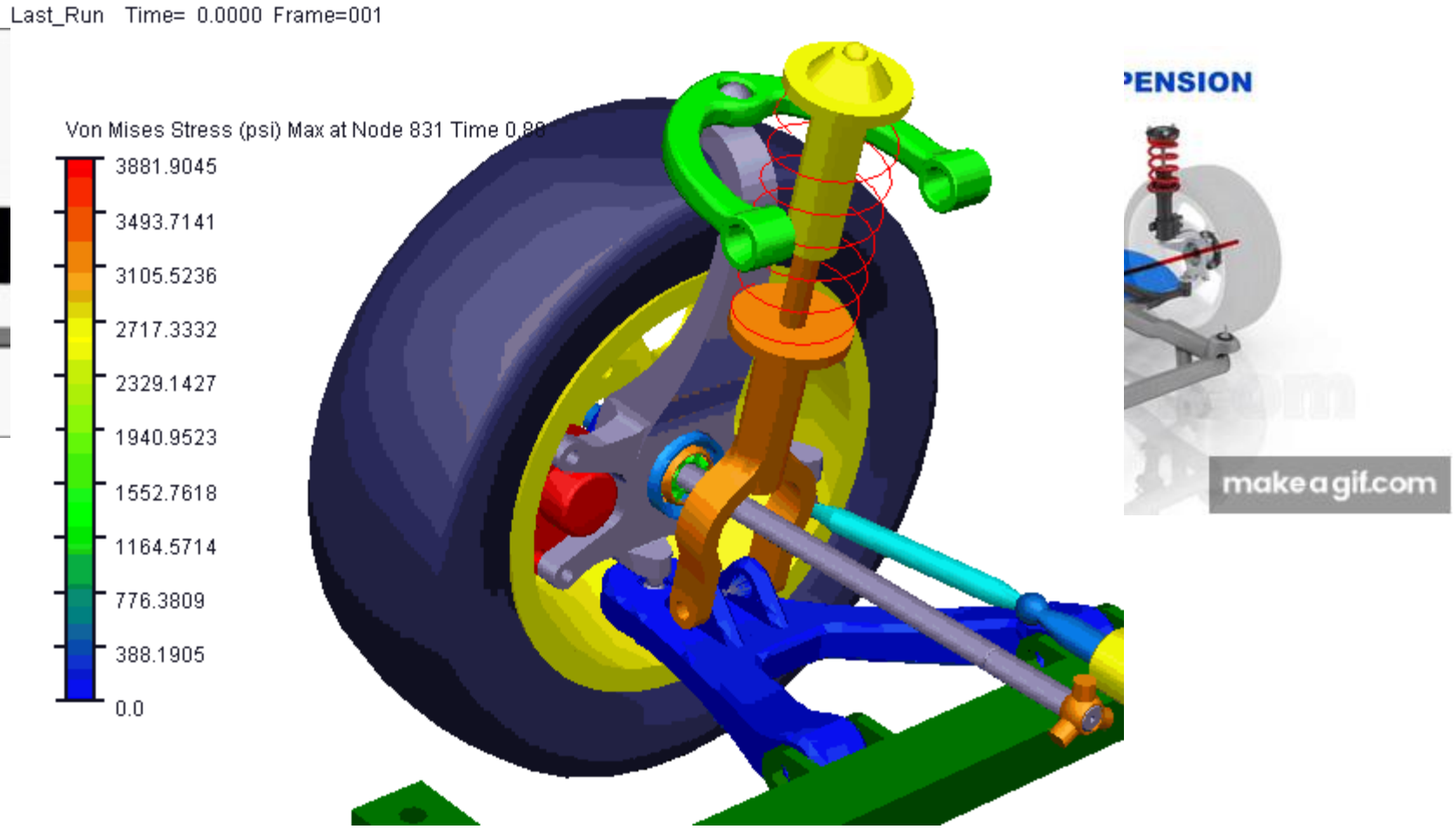

Rubber Band

Elastic Potential Energy



Elastic Potential Energy





Elastic Potential Energy

$$W_F = \mathrm{PE} = \frac{1}{2}\mathrm{K}x_2^2 - \frac{1}{2}\mathrm{K}x_1^2$$

$$W_e = -W_F$$

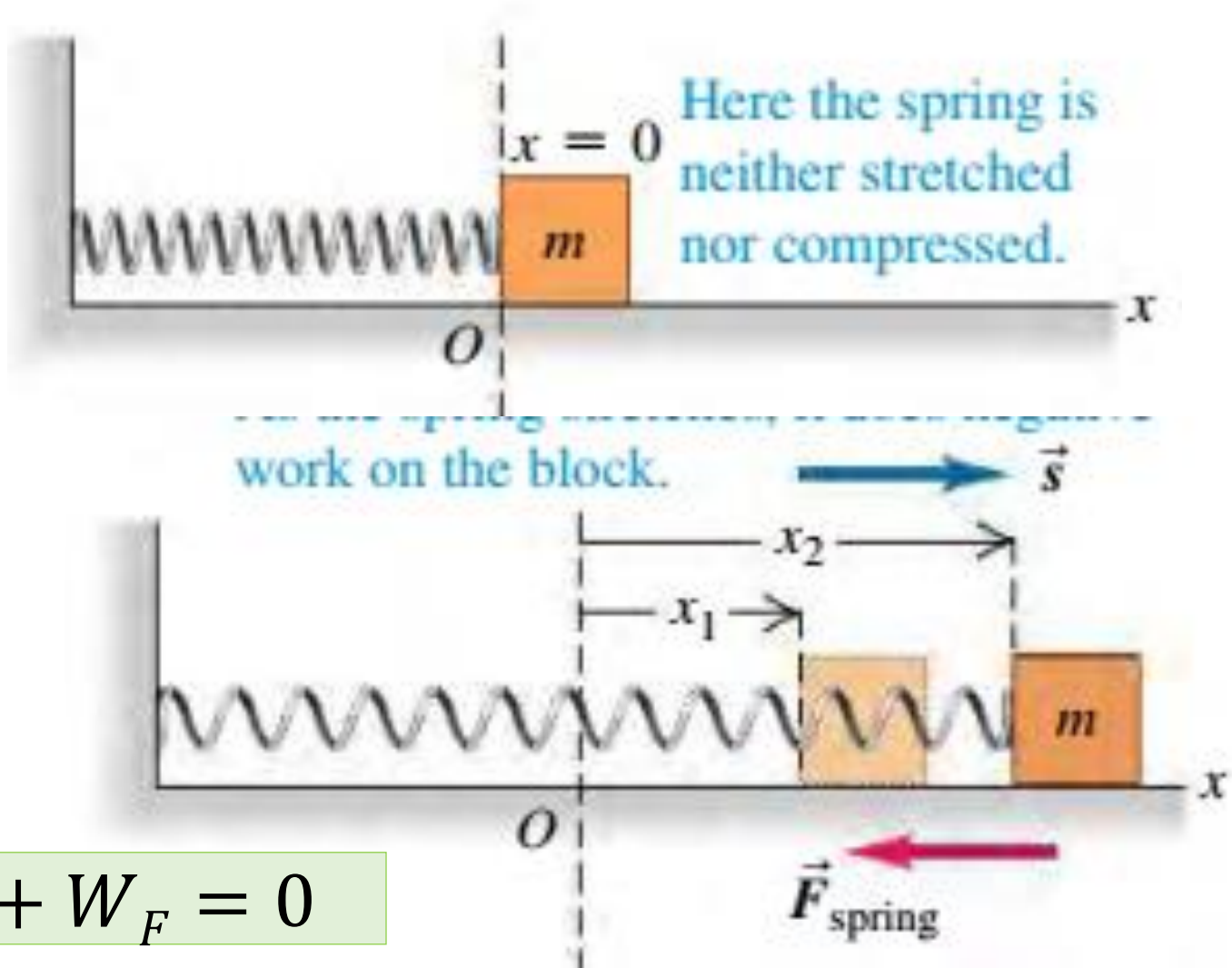
$$W_e = -\frac{1}{2}\mathrm{K}x_2^2 + \frac{1}{2}\mathrm{K}x_1^2$$

$$W_e = \frac{1}{2}\mathrm{K}x_1^2 - \frac{1}{2}\mathrm{K}x_2^2$$

$$W_e = U_{e1} _ Ue_2$$

$$W_e = -(U_{e2} - U_{e1})$$

$$W_e + W_F = 0$$



A glider with mass $m = 0.200$ kg sits on a frictionless, horizontal air track, connected to a spring with force constant $k = 5.00$ N/m. You pull on the glider, **stretching the spring 0.100 m**, and release it from rest. The glider moves back toward its equilibrium position ($x = 0$). **What is its *x*-velocity** when $x = 0.080$ m?

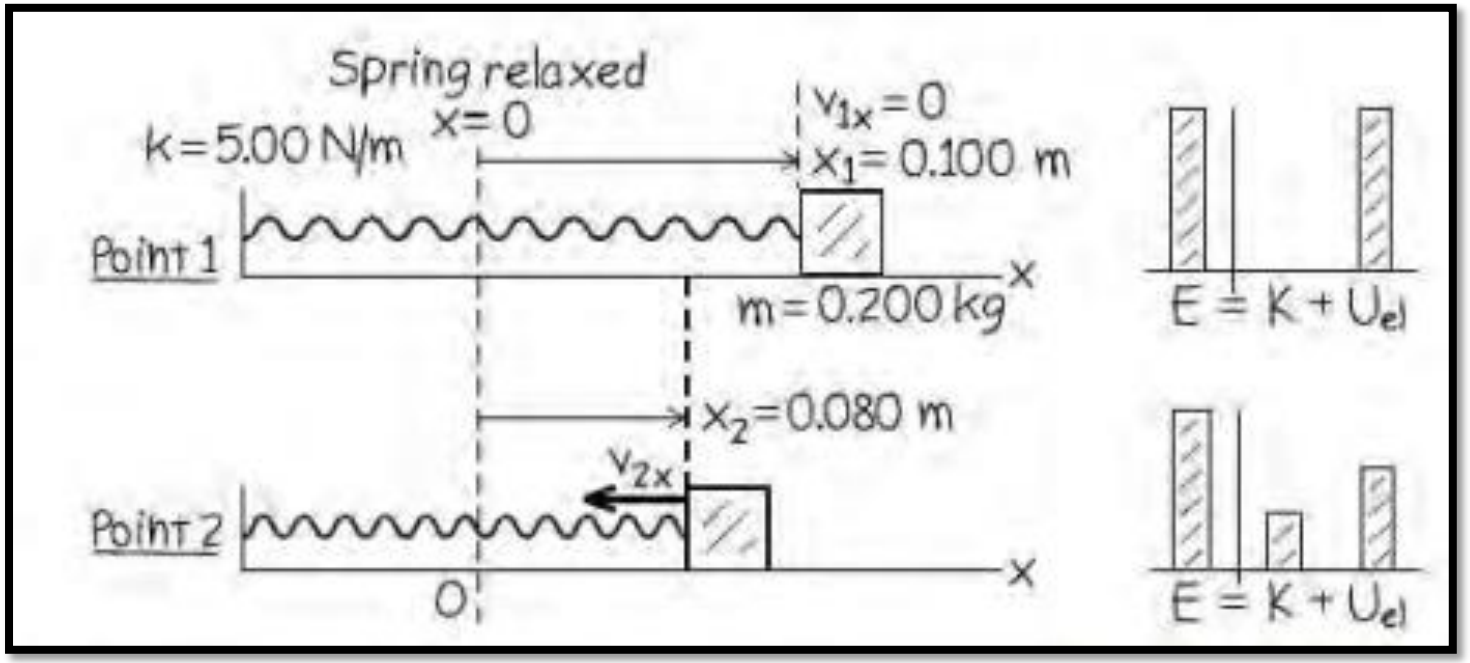
$V_{2x} = ?$

$$K_1 + U_1 = K_2 + U_2$$

$$K_2 = U_2 - (K_1 + U_1)$$

$$K_1 = \frac{1}{2}mv_{1x}^2 \qquad K_2 = \frac{1}{2}mv_{2x}^2$$

$$U_1 = \frac{1}{2}Kx_1^2 \qquad U_2 = \frac{1}{2}Kx_2^2$$



A glider with mass ***m* = 0.200 kg** sits on a frictionless, horizontal air track, connected to a spring with force constant ***k* = 5.00 N/m**. You pull on the glider, **stretching the spring 0.100 m**, and release it from rest. The glider moves back toward its equilibrium position ($x = 0$). **What is its *x*-velocity** when $x = 0.080$ m?

$$K_1 = \frac{1}{2}mv_{1x}^2 = \frac{1}{2}(0.2)(0)_{1x}^2 = 0\ J$$

$$U_1 = \frac{1}{2}Kx_1^2 = \frac{1}{2}(5)(0.1)_1^2 = 0.025\ J$$

$$K_2 = \frac{1}{2}mv_{2x}^2$$

$$U_2 = \frac{1}{2}Kx_2^2 = \frac{1}{2}(5)(0.08)_2^2 = 0.016\ \text{J}$$

Negative root because the glider is moving in the –x direction.

A glider with mass $m = 0.200$ kg sits on a frictionless, horizontal air track, connected to a spring with force constant $k = 5.00$ N/m. You pull on the glider, **stretching the spring 0.100 m**, and release it from rest. The glider moves back toward its equilibrium position ($x = 0$). **What is its *x*-velocity** when $x = 0.080$ m?

$$K_2 = U_2 - (K_1 + U_1)$$

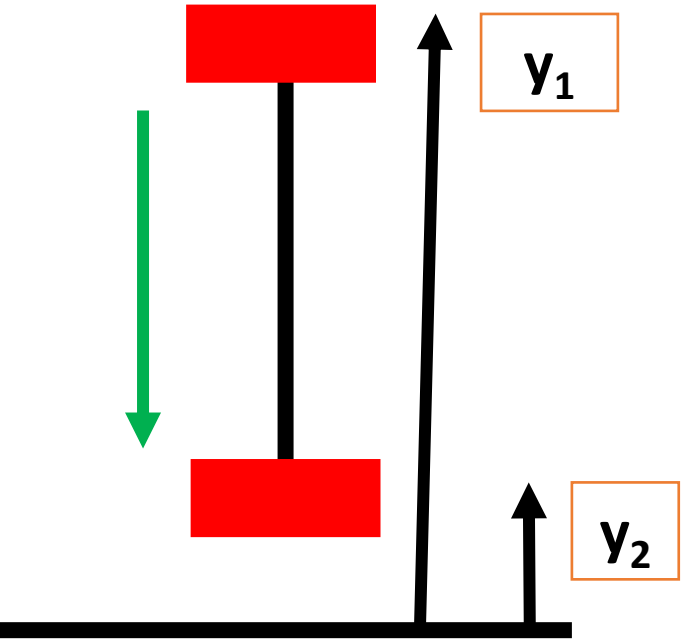
$$K_2 = \frac{1}{2} m v_{2x}^2$$

$$v_{2x} = \pm\sqrt{\frac{2K_2}{m}} = \pm\sqrt{\frac{2(0.009)}{0.2}} = \pm 0.3\ m/s$$

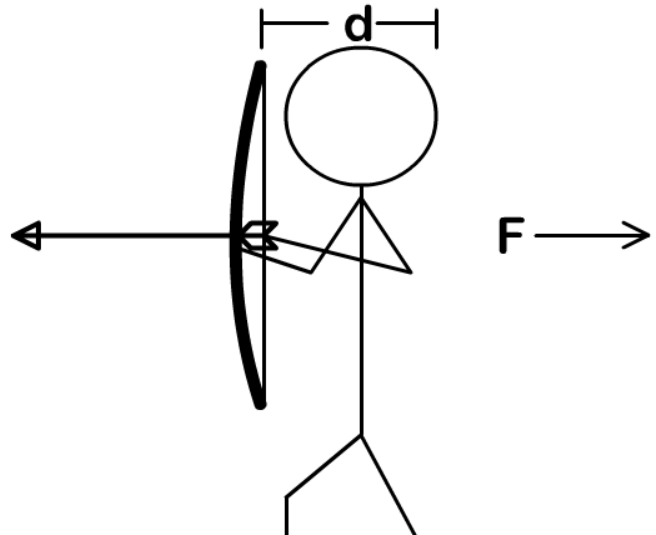
Negative root because the glider is moving in the –x direction.

Potential Energy

Potential energy is energy an object or system possesses due to its position, arrangement, or state. It has the *potential* to be converted into kinetic energy.



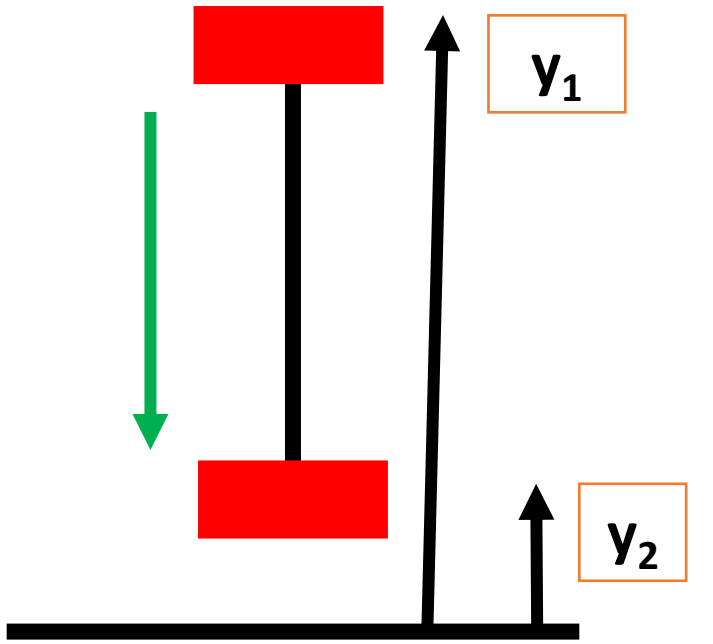
$$W = (KE)_f - (KE)_i$$



Potential Energy

-ve sign; When the object moves up, y increases, the work done by the gravitational force is negative, and the gravitational potential energy increases

What is work done by gravity?



$$W = \boldsymbol{F}.\boldsymbol{d}$$

$$W = Fd = mgh$$

$$W = mg(y_1 - y_2)$$

$$W = mgy_1 - mgy_2$$

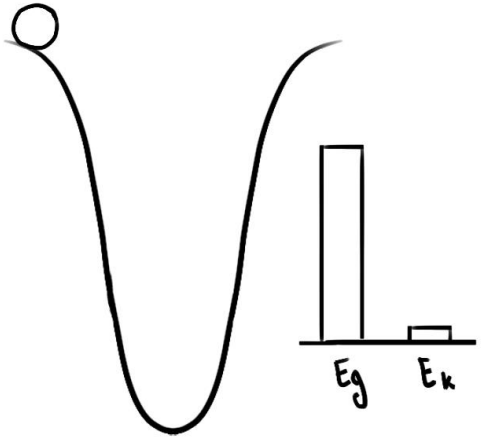
$$W = U_1 - U_2$$

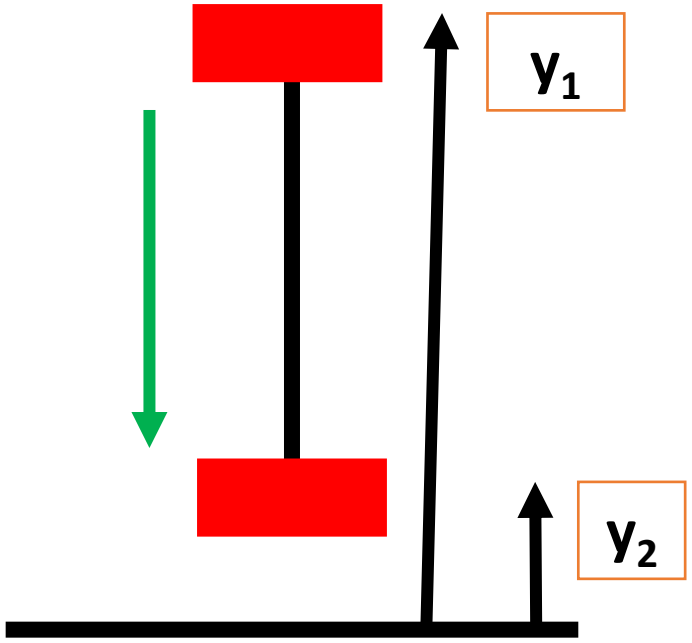
$$W = -(U_2 - U_1)$$

$$W = -\Delta U$$

$$U = mgh$$

Conversion of Mechanical Energy





$$W = K_2 - K_1$$

$$W = -(U_2 - U_1)$$

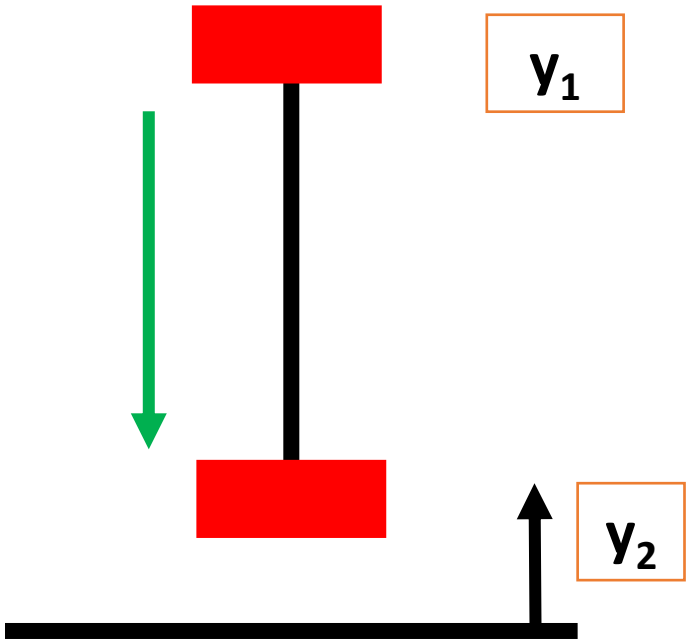
$$K_2 - K_1 = -(U_2 - U_1)$$

$$K_2 - K_1 = -U_2 + U_1$$

$$K_2 + U_2 = K_1 + U_1$$

Energy is conserved

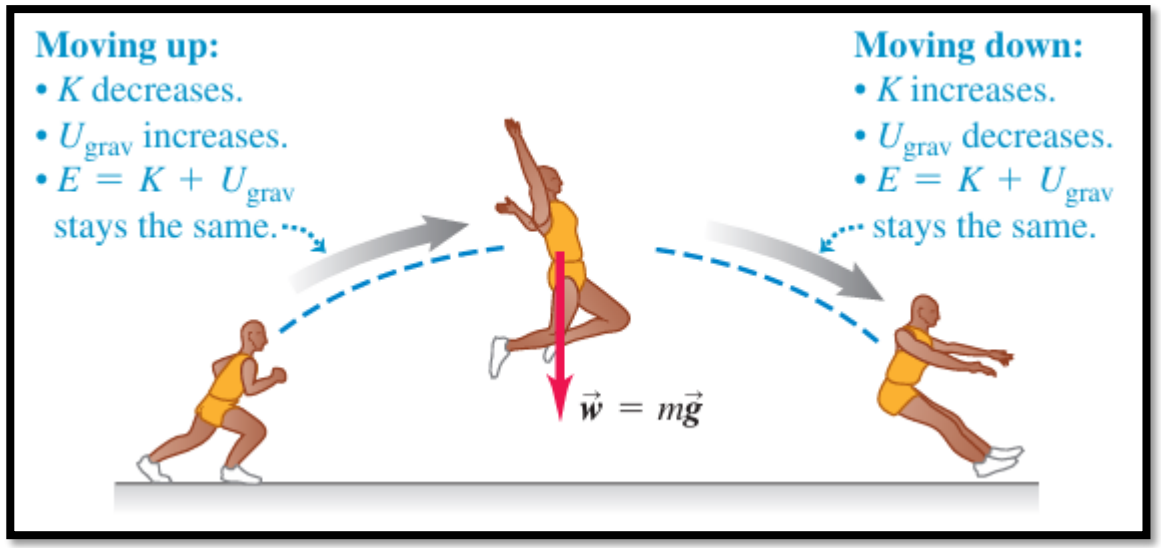
Conversion of Mechanical Energy



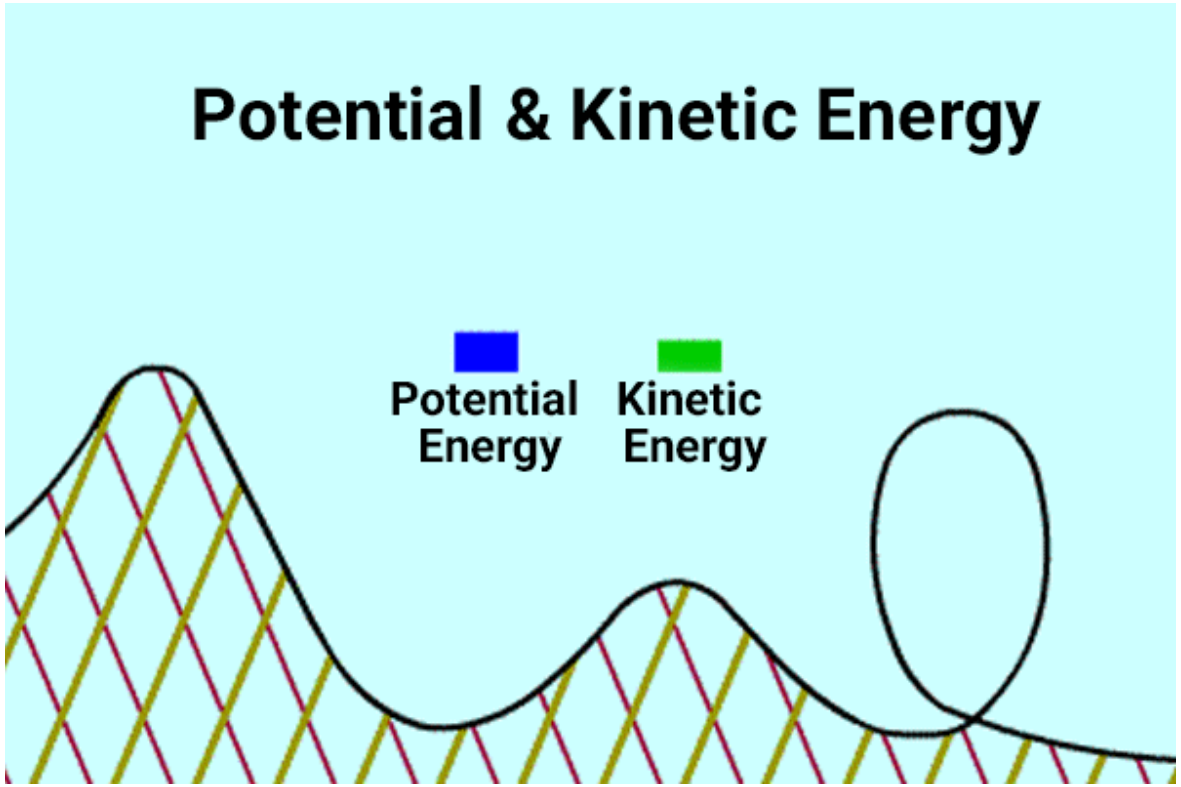
$$K_2 + U_2 = K_1 + U_1$$

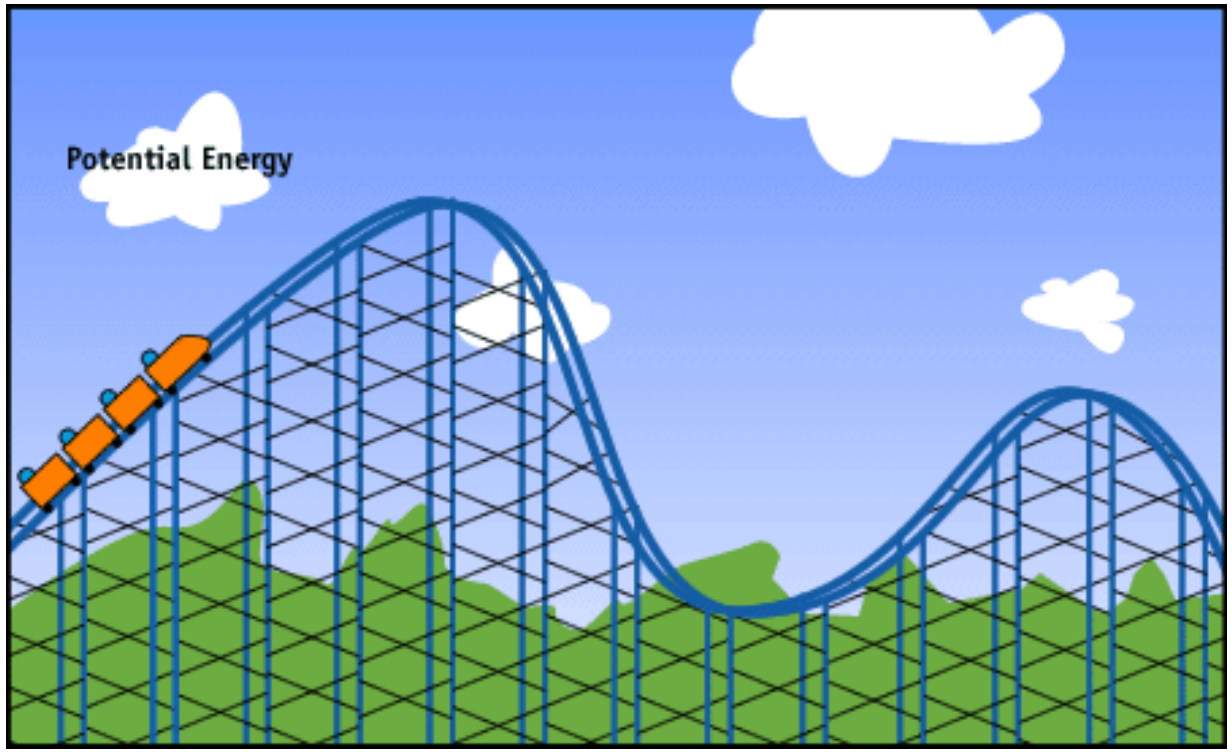
$$\mathrm{E} = K + U = Constant$$

True only “gravity” does work



Conversion of Mechanical Energy





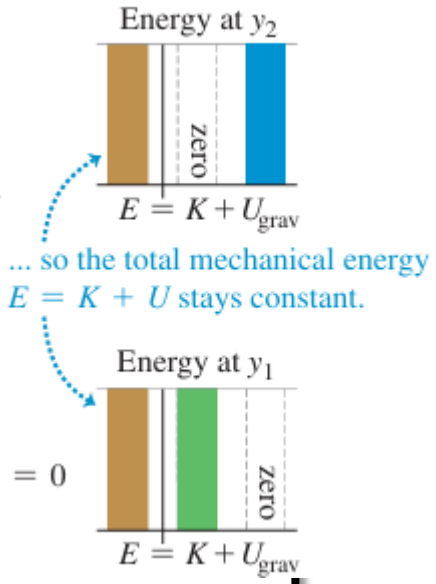
Example

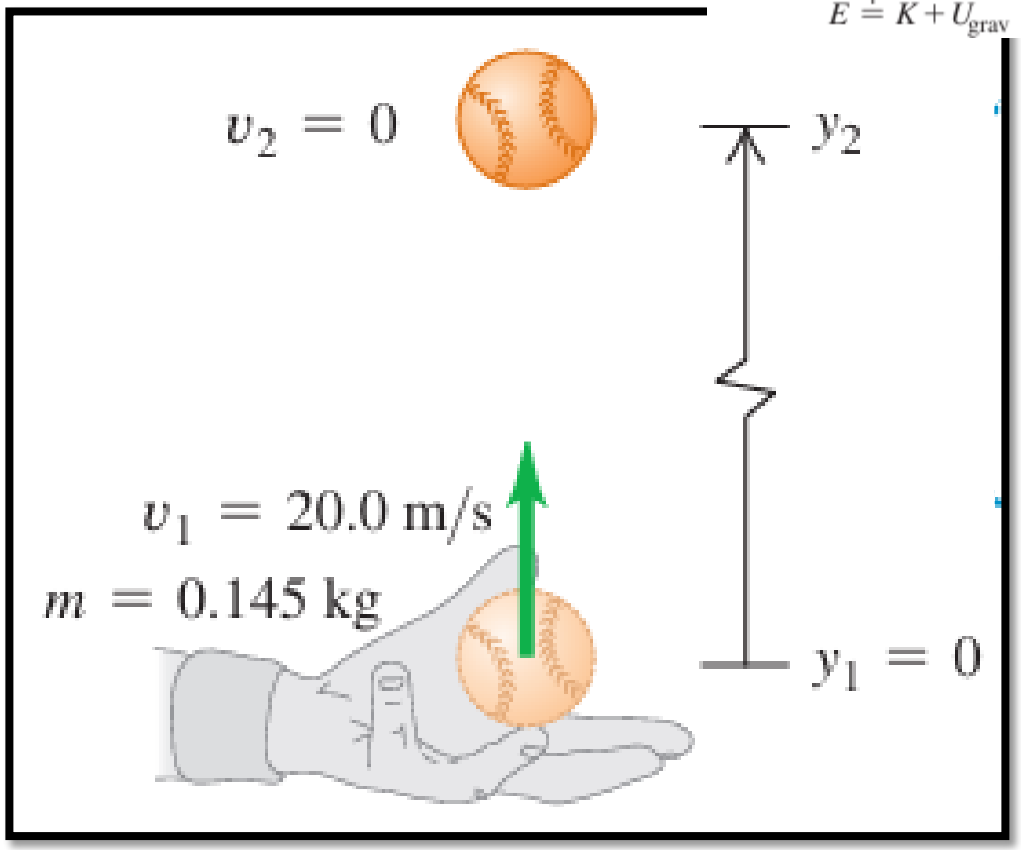
You throw a **0.145-kg** baseball straight up, giving it an **initial velocity of ma**
20.0 m/s. **Find how high** it goes, ignoring air resistance.

$$K_2 + U_2 = K_1 + U_1$$

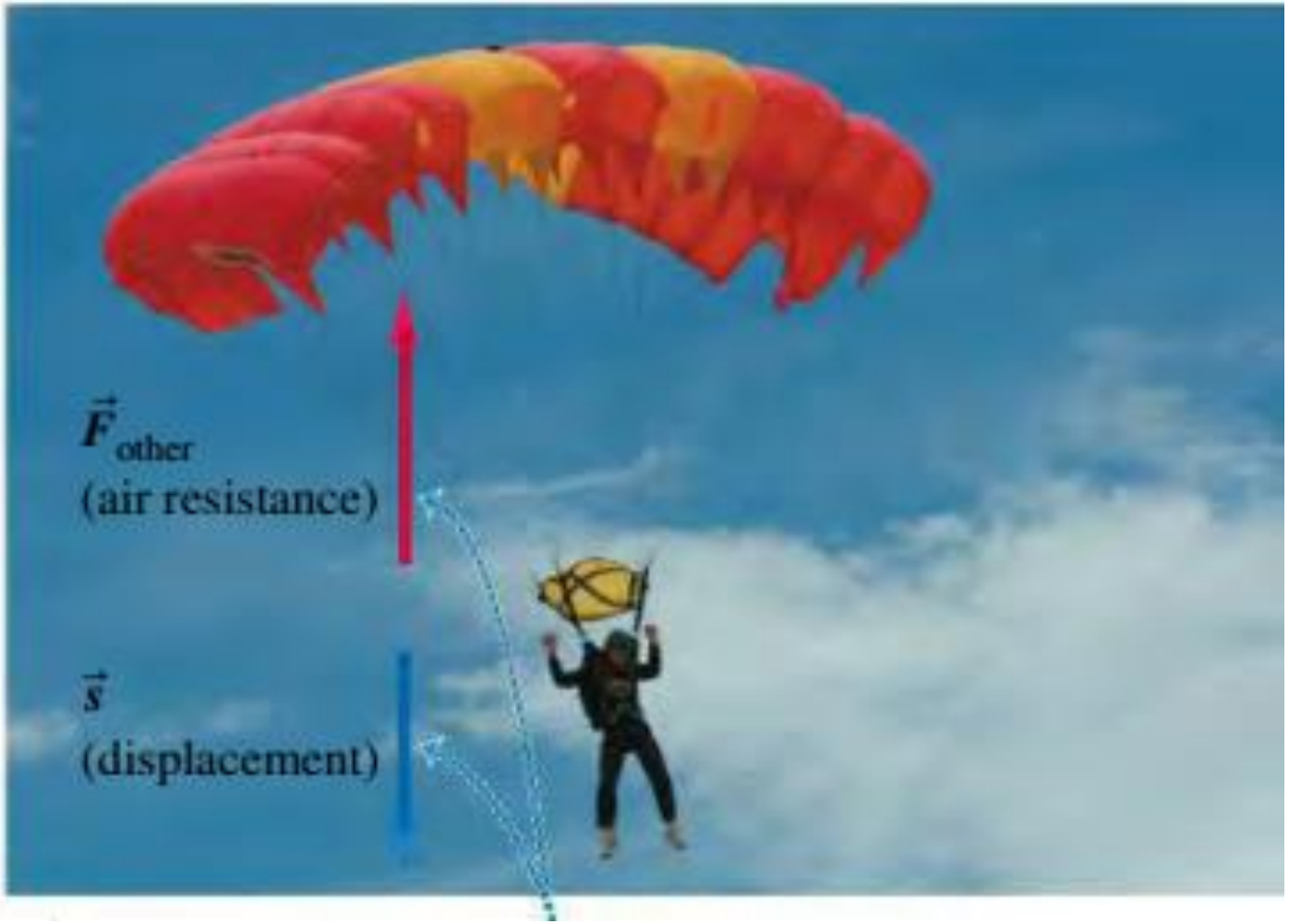
Initial	Final
$KE_1 = \frac{1}{2}mV_1^2$	$KE_2 = 0$
$U_1 = 0$	$U_2 = mgy_2$

$$y_2 = 20.4\ \text{m}$$

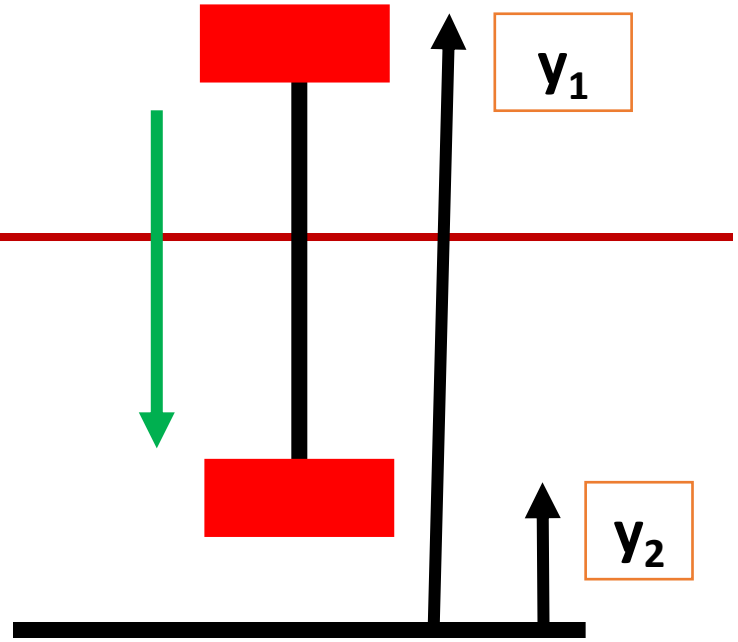




Conversion of Mechanical Energy



Elastic Potential Energy



$$K_1 + U_1 = K_2 + U_2$$

$$K_1 + U_{g1} = K_2 + U_{g2}$$

$$K_1 + U_{e1} = K_2 + U_{e2}$$

$$K_1 + U_{g1} + U_{e1} = K_2 + U_{g2} + U_{e2}$$

$$W_{other} + K_1 + U_{g1} + U_{e1} = K_2 + U_{g2} + U_{e2}$$

